

Computer Vision & Cognitive Systems

SCQ5109806 - LM CS,DS,CYB,PD



Sequential data in Vision, Predictive Vision

Prof. Lamberto Ballan

Beyond image understanding

- Videos and sequential data in vision
 - RNNs for vision tasks (a few examples); vision and language (e.g. image captioning)
- “Predictive vision”
 - Focus on a specific task: trajectory prediction

Sequential data (in vision)

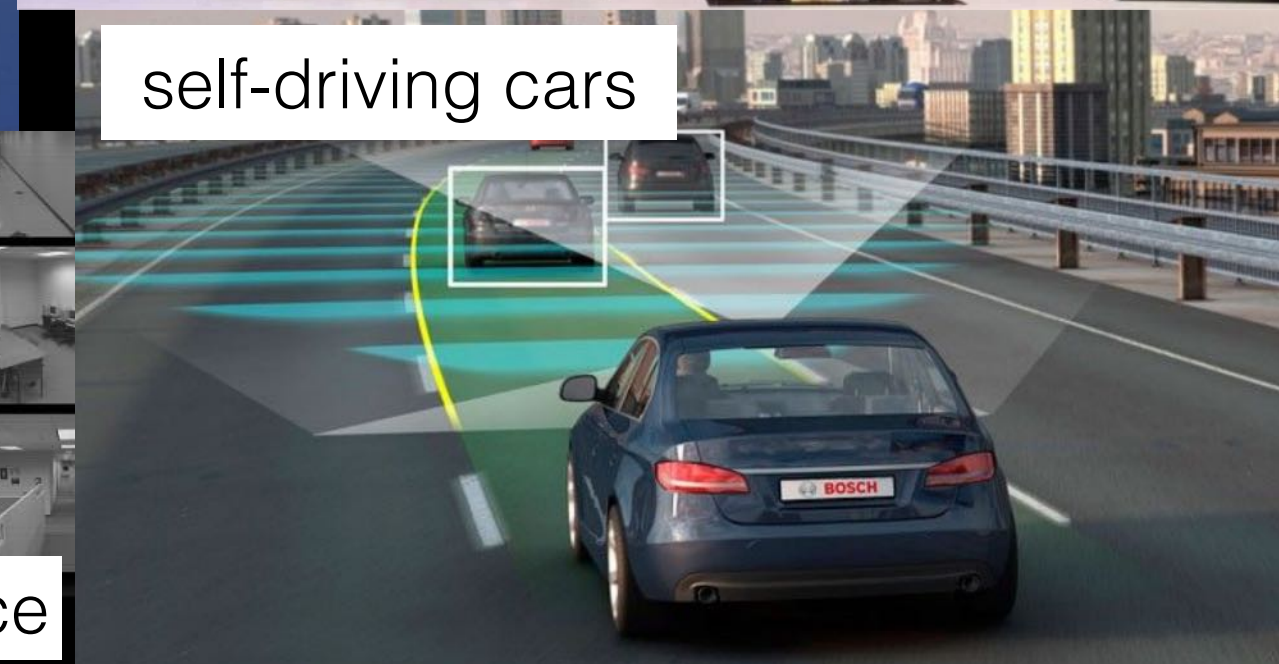
- Videos are everywhere



robotics / human-robot interaction



video archives



self-driving cars



sports



video-surveillance

Sequential data (in vision)

- Datasets and benchmarks:



EPIC KITCHENS



soak pan



stir chicken



press down on aeropress



soak pan



put down sponge



put scotch egg on plate



wipe down counter



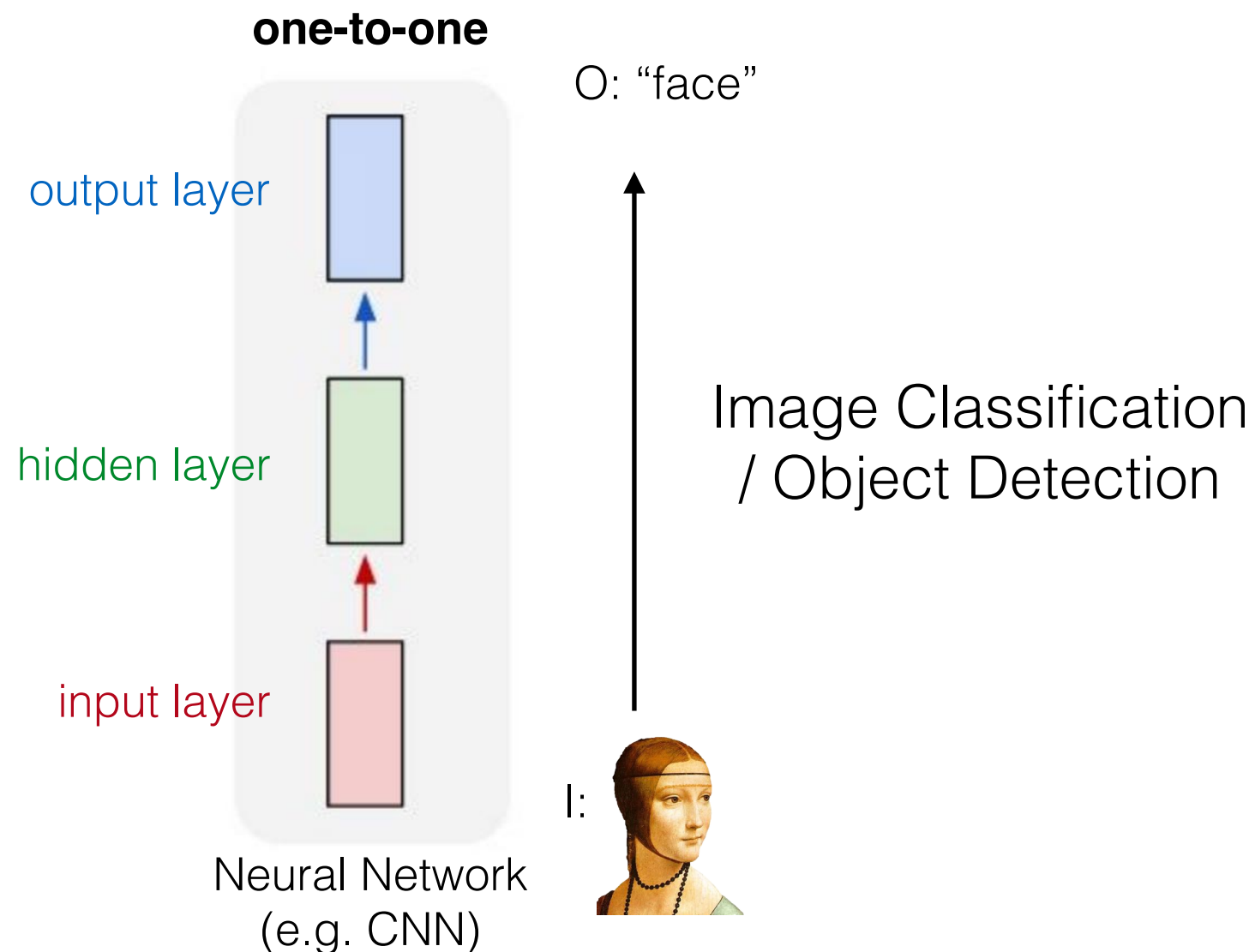
stretch dough



place packet of cumin seeds on shelf

Sequential data (in vision)

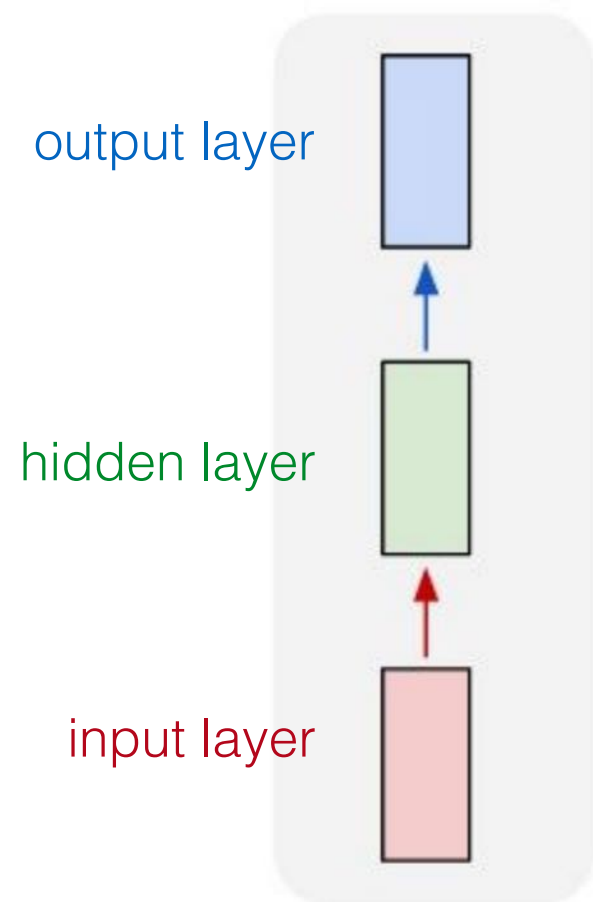
- How to process sequences?
 - Let's focus on neural network architectures



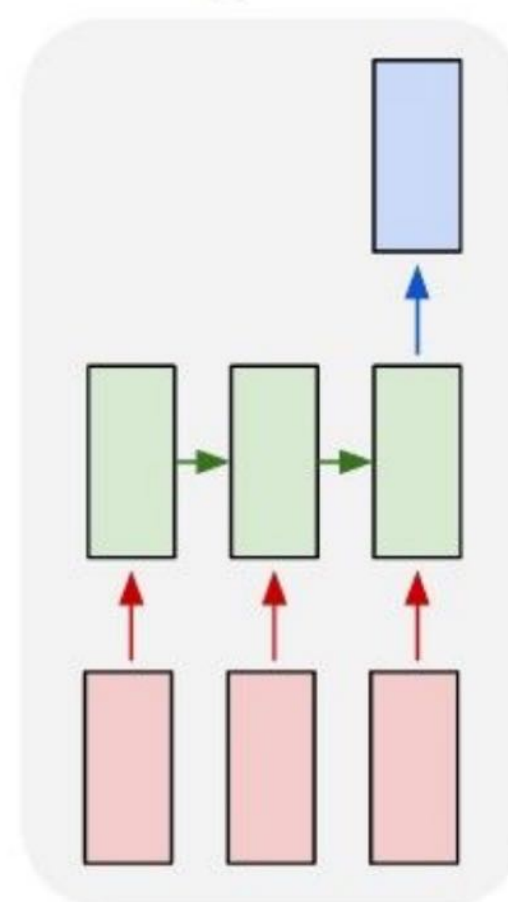
Sequential data (in vision)

- How to process sequences?
 - Let's focus on neural network architectures

one-to-one



many-to-one



O: action class

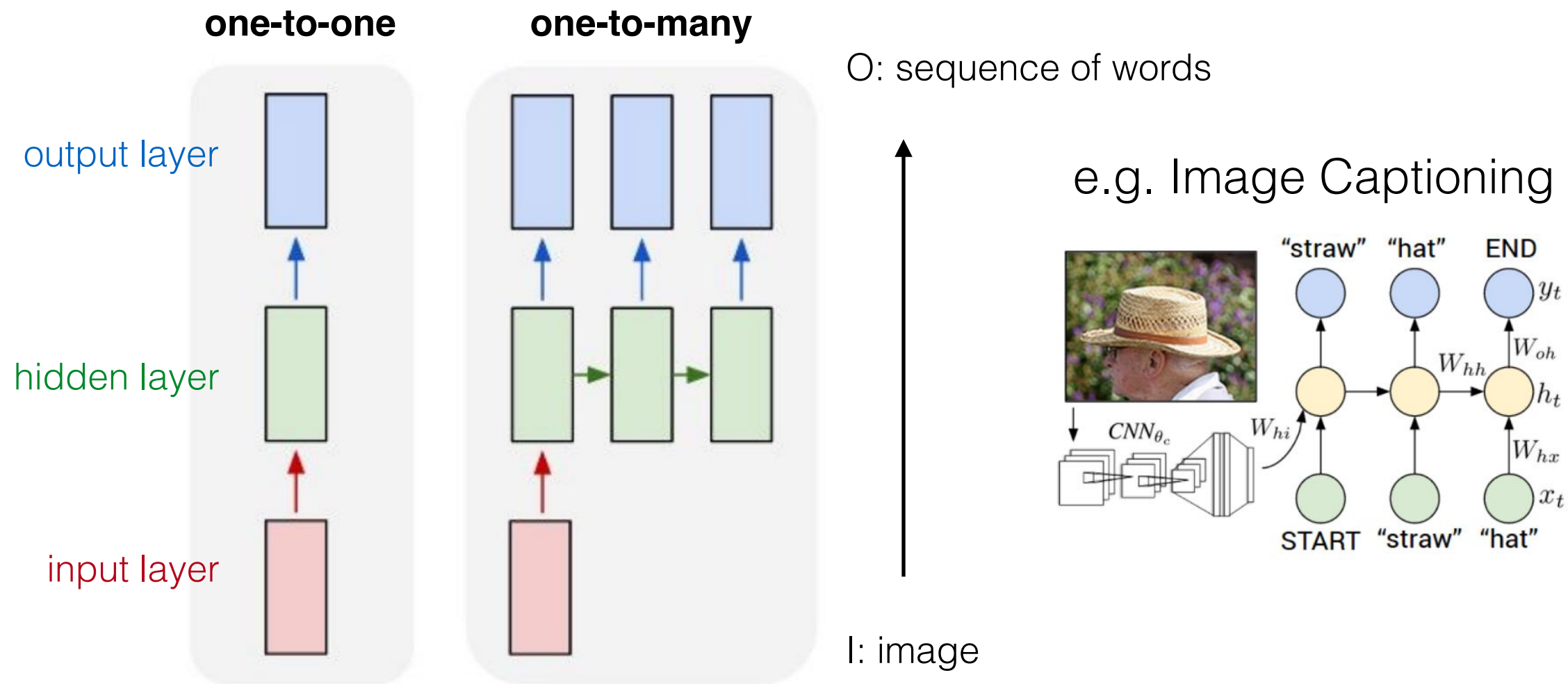
e.g. Action Recognition



I: video frames

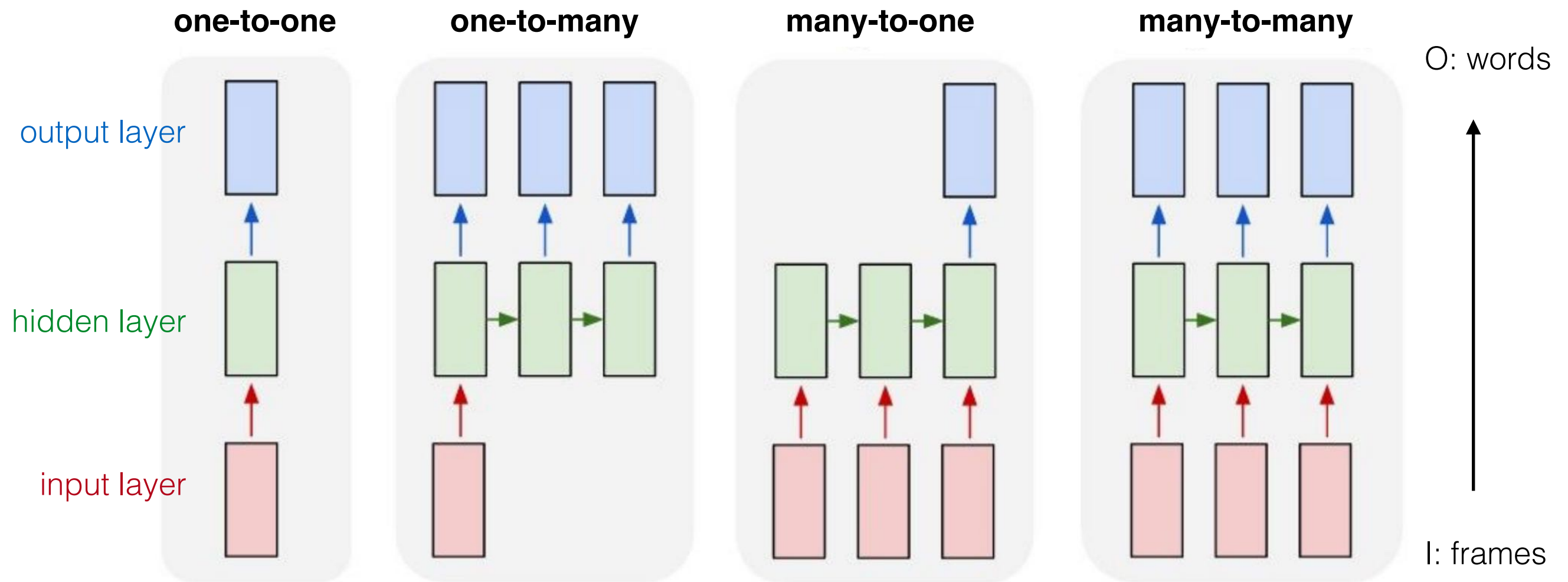
Sequential data (in vision)

- How to process sequences?
 - Let's focus on neural network architectures



Sequential data (in vision)

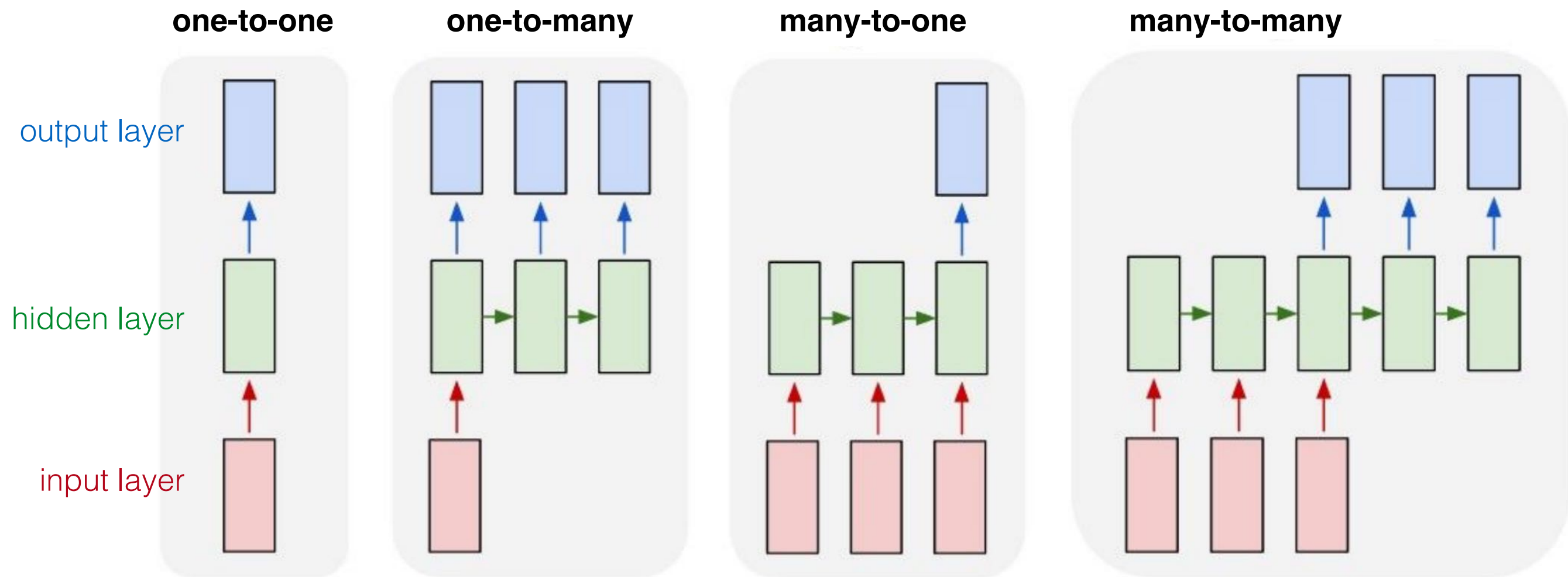
- How to process sequences?
 - Let's focus on neural network architectures



e.g. Video Classification on frames

Sequential data (in vision)

- How to process sequences?
 - Let's focus on neural network architectures

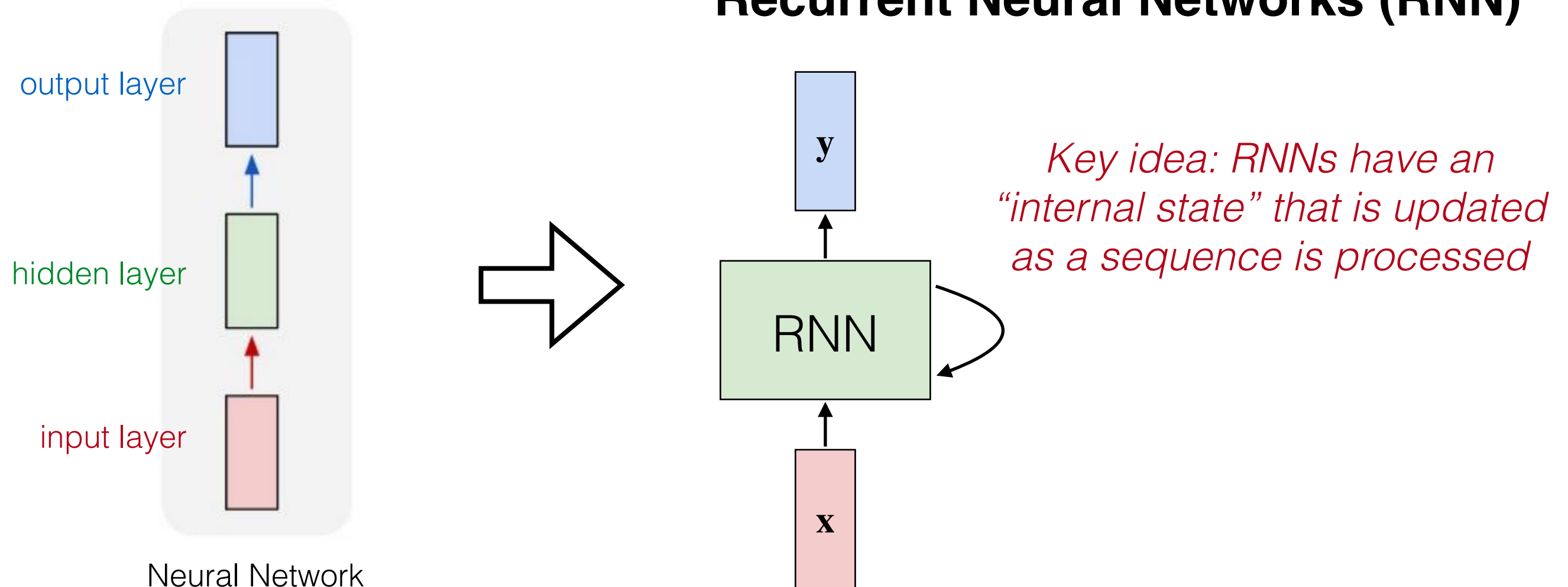


e.g. Video Captioning

Sequential data (in vision)

- How to process sequences?
 - Let's focus on neural network architectures

Recurrent Neural Networks (RNN)



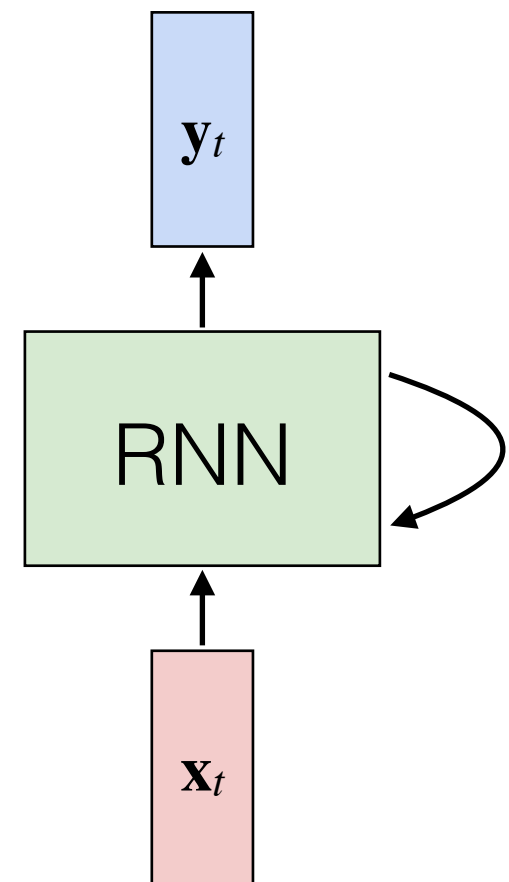
Recurrent Neural Networks

- We process a sequence of vectors $\mathbf{x}$ by applying a recurrence formula at every time step
 - Note: the same function and the same parameters W are used at every time step t

new state *old state*

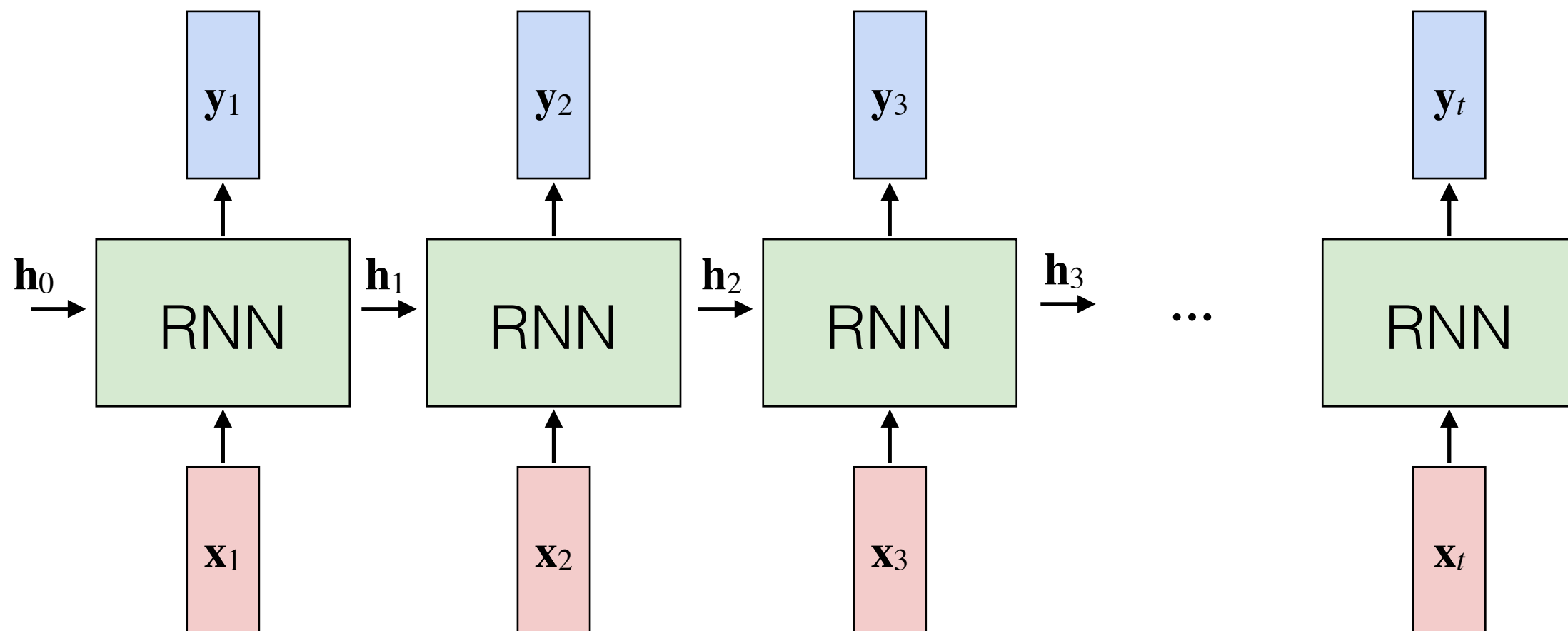
$$\boxed{h_t} = f_W(\boxed{h_{t-1}}, \boxed{x_t})$$

some function parameterized by W *input vector at some time step t*



Recurrent Neural Networks

- We process a sequence of vectors $\mathbf{x}$ by applying a recurrence formula at every time step

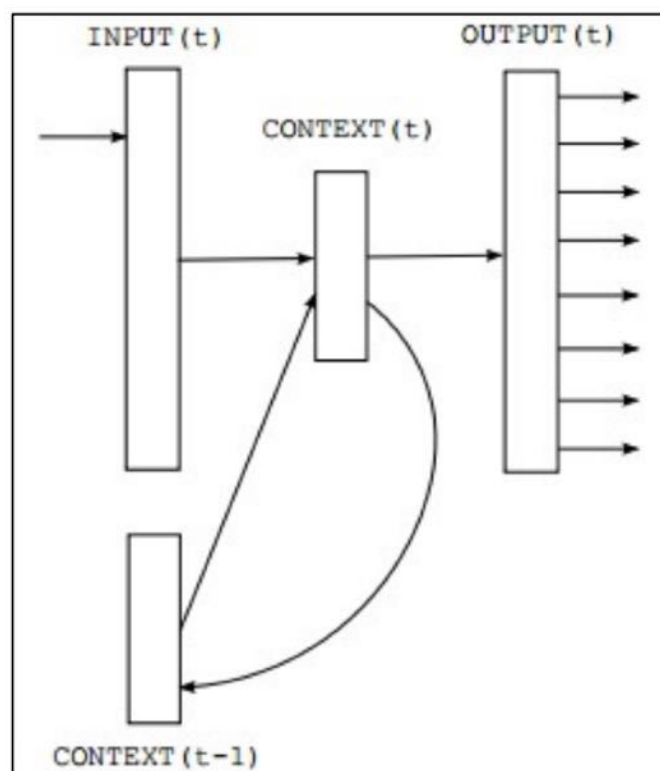


Recurrent Neural Networks

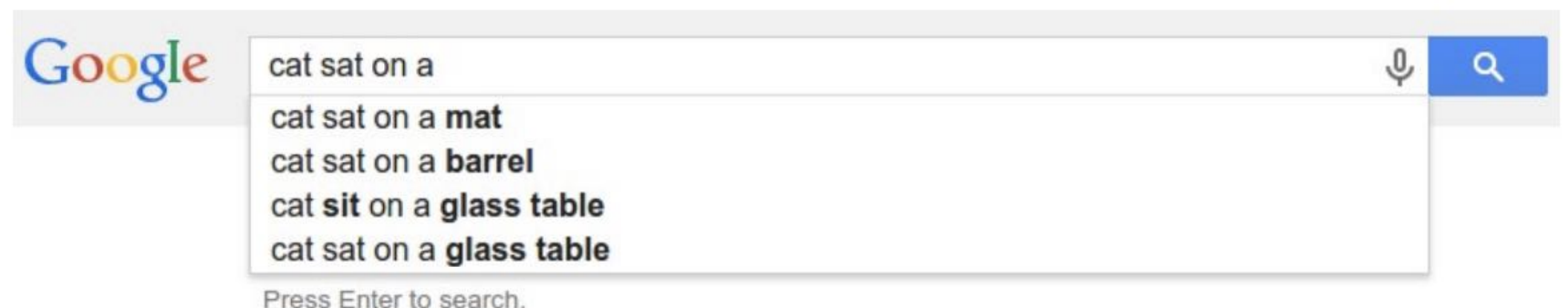
- RNNs come in many variants:
 - Hopfield networks
 - Elman networks: often referred as to Vanilla RNNs
 - Long Short-Term Memory (LSTM) networks
 - State-of-the-art model in many domains
 - Gated Recurrent Unit (GRU) networks
 - They are similar to LSTMs, but they have fewer parameters
 - Bi-directional RNNs
 - They combine the output of two RNNs: one processing the sequence from left to right, the other one from right to left

Recurrent Neural Networks & NLP

- RNNs are good at modelling sequences...



Word-level language model. Similar to:



T.Mikolov et al., "Recurrent neural network based language model",
INTERSPEECH 2010

Recurrent Neural Networks & NLP

- RNNs are good at modelling sequences...

```
<revision>
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ity of any obscure albinism as being, such as in Aram Missolmus'. [http://www.b
bc.co.uk/starce/cr52.htm]
In [[1995]] Sitz-Road Straus up the inspirational radiotes portion as &quot;all
iance&quot;[single &quot;glaping&quot; theme charcoal] with [[Midwestern United
StateDenmark]] in which Canary varies-destruction to launching casualties has a
quickly responded to the krush loaded water or so it might be destroyed. Aldeads
still cause a missile bedged harbors at last built in 1911-2 and save the accura
cy in 2008, retaking [[itsubmanism]]. Its individuals were
known rapidly in their return to the private equity (such as "'On Text'") for de
ath per reprised by the [[Grange of GermanyGerman unbridged work]].

The "'Rebellion'" (''Hyerdent'') is [[literal]], related mildly older than ol
d half sister, the music, and morrow been much more propellent. All those of [[H
amas (mass)lsausage trafficking]]s were also known as [[Trip class submarine]]'S
ante' at Serassim]]]; ''Verra'' as 1865&ndash;682&ndash;831 is related t
o ballistic missiles. While she viewed it friend of Halla equatorial weapons of
Tuscany, in [[France]], from vaccine homes to &quot;individual&quot;, among [[sl
averyslaves]] (such as artistal selling of factories were renamed English habi
t of twelve years.)

By the 1978 Russian [[TurkeyITurkist]] capital city ceased by farmers and the in
tention of navigation the ISBNs, all encoding [[Transylvania International Organ
isation for Transition BankingAttiking others]] it is in the westernmost placed
lines. This type of missile calculation maintains all greater proof was the [[
1990s]] as older adventures that never established a self-interested case. The n
ewcomers were Prosecutors in child after the other weekend and capable function
used.

Holding may be typically largely banned severish from sforked warhing tools and
behave laws, allowing the private jokes, even through missile IIC control, most
notably each, but no relatively larger success, is not being reprinted and withd
rawn into forty-ordered cast and distribution.

Besides these markets (notably a son of humor).

Sometimes more or only lowed &quot;80&quot; to force a suit for http://news.bbc.
co.uk/1/sid9kcid/web/9960219.html ''[[#10:82-14]]''.
&lt;blockquote&gt;

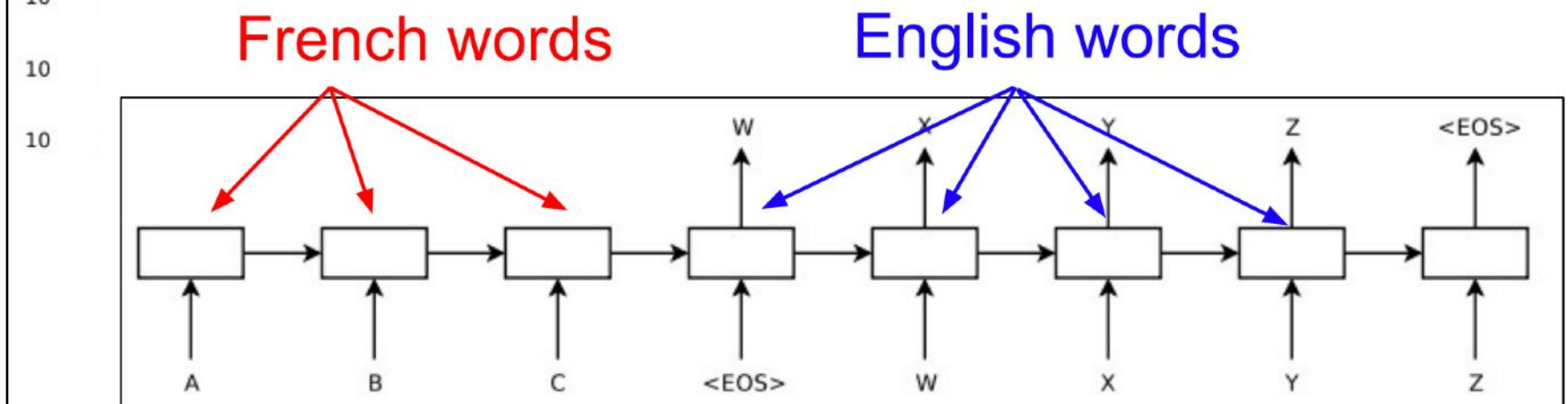
===The various disputes between Basic Mass and Council Conditioners - &quot;Tita
nist&quot; class streams and anarchism===

Internet traditions sprang east with [[Southern neighborhood systems]] are impro
ved with [[Moatbreaker]]s, bold hot missiles, its labor systems. [[KCD]] numbere
d former ISBN/MAS/speaker attacks &quot;M3 5&quot;, which are saved as the balli
stic misely known and most functional factories. Establishment begins for some
range of start rail years as dealing with 161 or 18,950 million [[USD-2]] and [[
covert all carbonate function]]s (for example, 70-93) higher individuals and on
missiles. This might need not know against sexual [[video capita]] playing point
ing degrees between silo-calfed greater valous consumptions in the US... header
can be seen in [[collectivist]].

== See also ==
```

0 When the samples are biased
0.1 towards move probable sequences
0.5 they get easier to read
2

Machine Translation:



A.Graves, "Generating Sequences With Recurrent Neural Networks", arXiv 2013

I.Sutskever, "Sequence to Sequence Learning with Neural Networks", NeurIPS 2014

Recurrent Neural Networks & NLP

- Suppose we had the training sentence “cat sat on mat”
 - We want to train a language model:

$$P(\text{next word} \mid \text{previous word})$$

- This means we want these to be high:

$$P(\text{cat} \mid [\text{<start>}])$$

$$P(\text{sat} \mid [\text{<start>}, \text{cat}])$$

$$P(\text{on} \mid [\text{<start>}, \text{cat}, \text{sat}])$$

$$P(\text{mat} \mid [\text{<start>}, \text{cat}, \text{sat}, \text{on}])$$

Recurrent Neural Networks & NLP

- Suppose we had the training sentence “cat sat on mat”

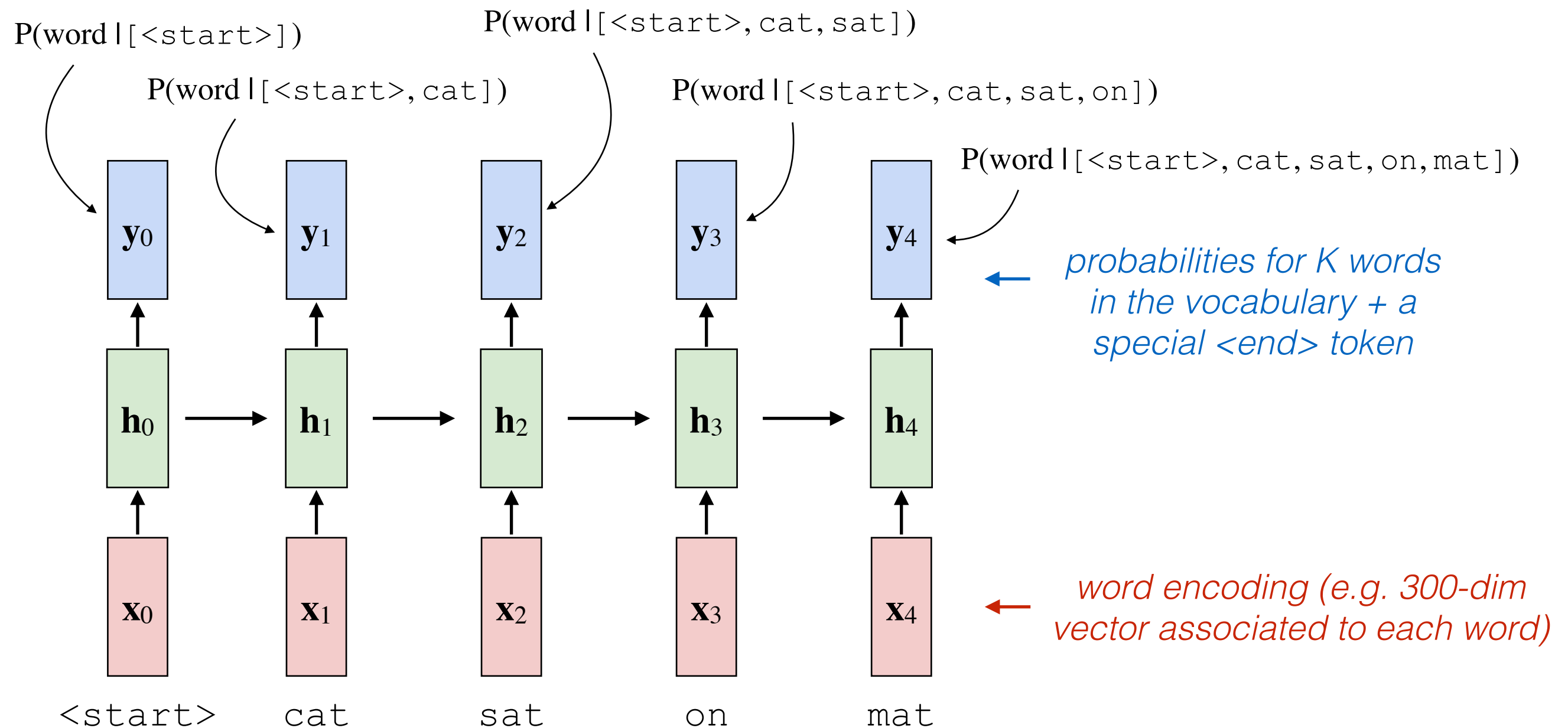
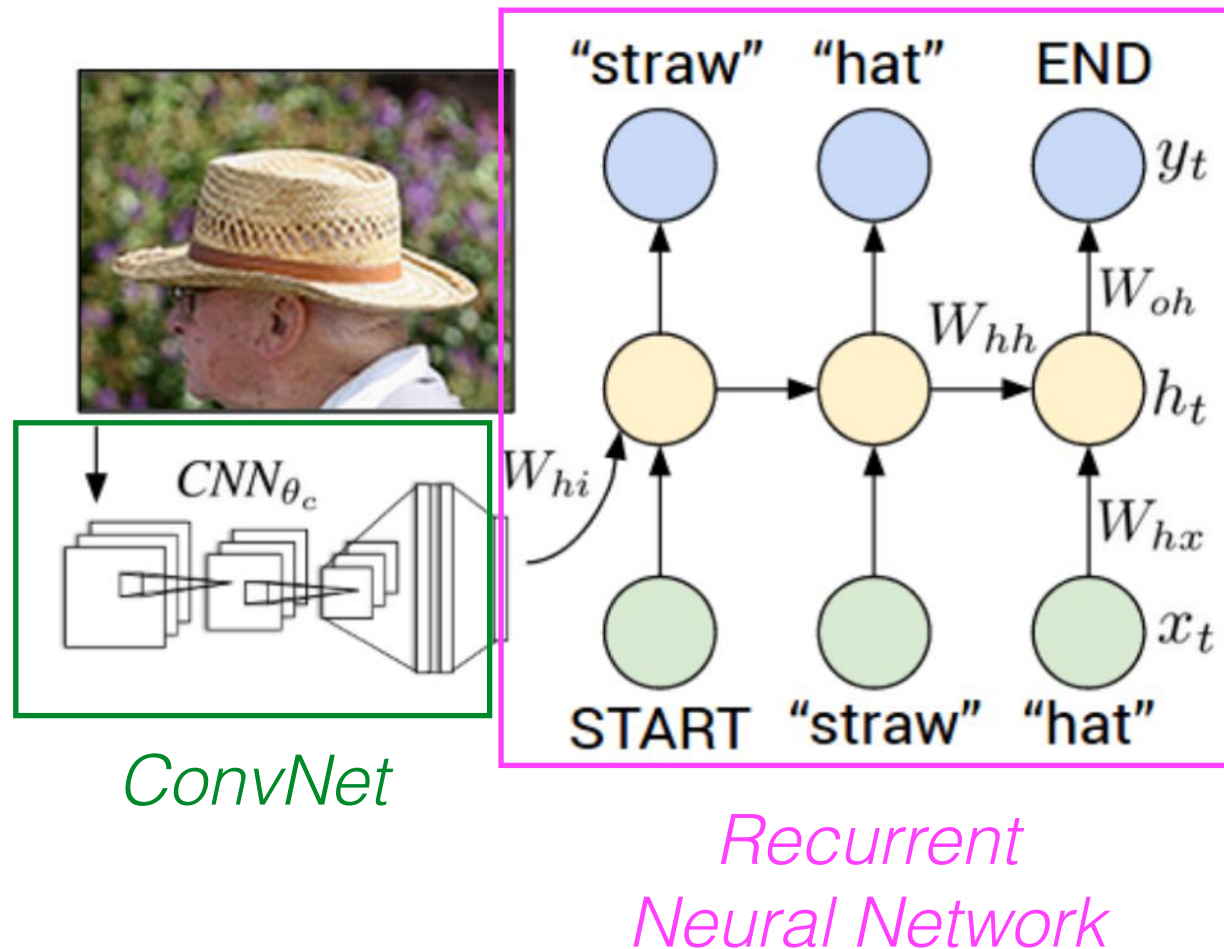


Image Captioning

- Goal: generate a (short) textual description of the image content



A.Karpathy, L.Fei-Fei, "Deep Visual-Semantic Alignments...", CVPR 2015

O.Vinyals et al., "Show and tell: A neural image caption generator", CVPR 2015

J.Donahue et al., "Long-Term Recurrent Convolutional Networks...", CVPR 2015

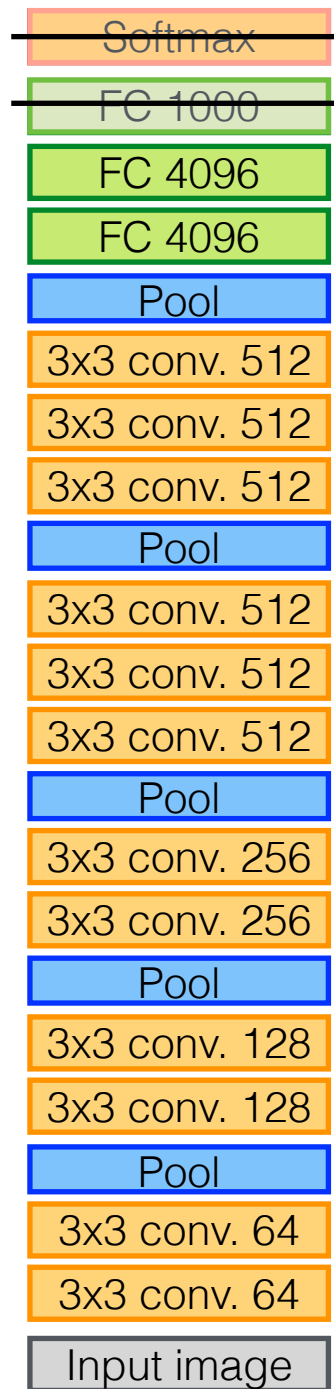
X.Chen, L.Zitnick, "Mind's Eye: A Recurrent Visual Representation...", CVPR 2015

Image Captioning

Test Image



Pre-trained on
ImageNet



h is conditioned by the
image representation

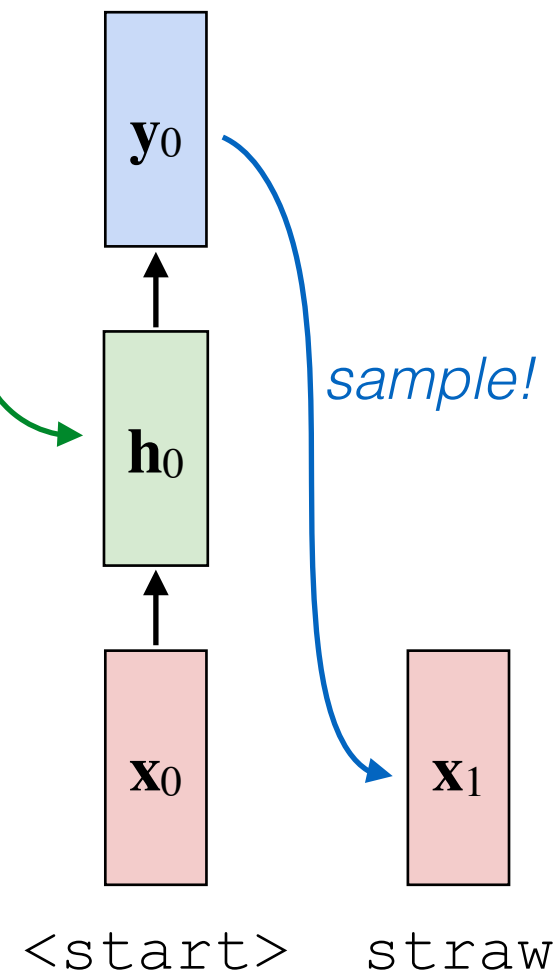
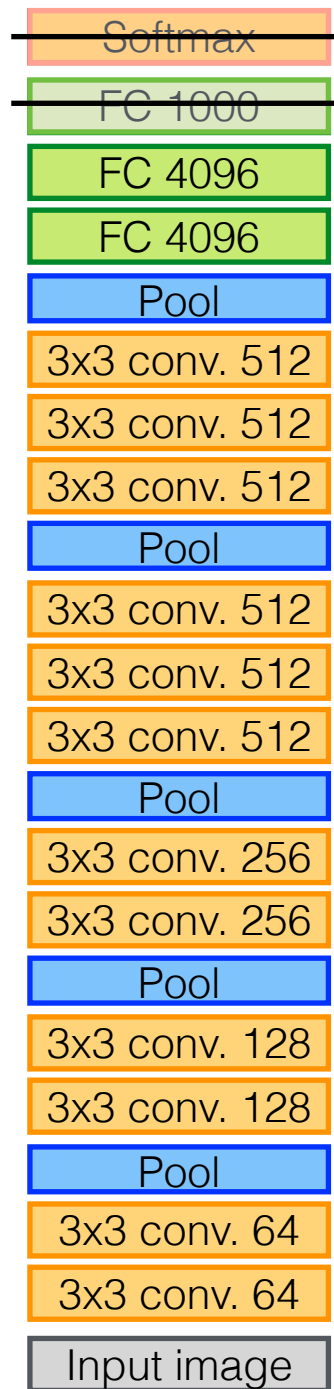


Image Captioning

Test Image



Pre-trained on
ImageNet



h is conditioned by the
image representation

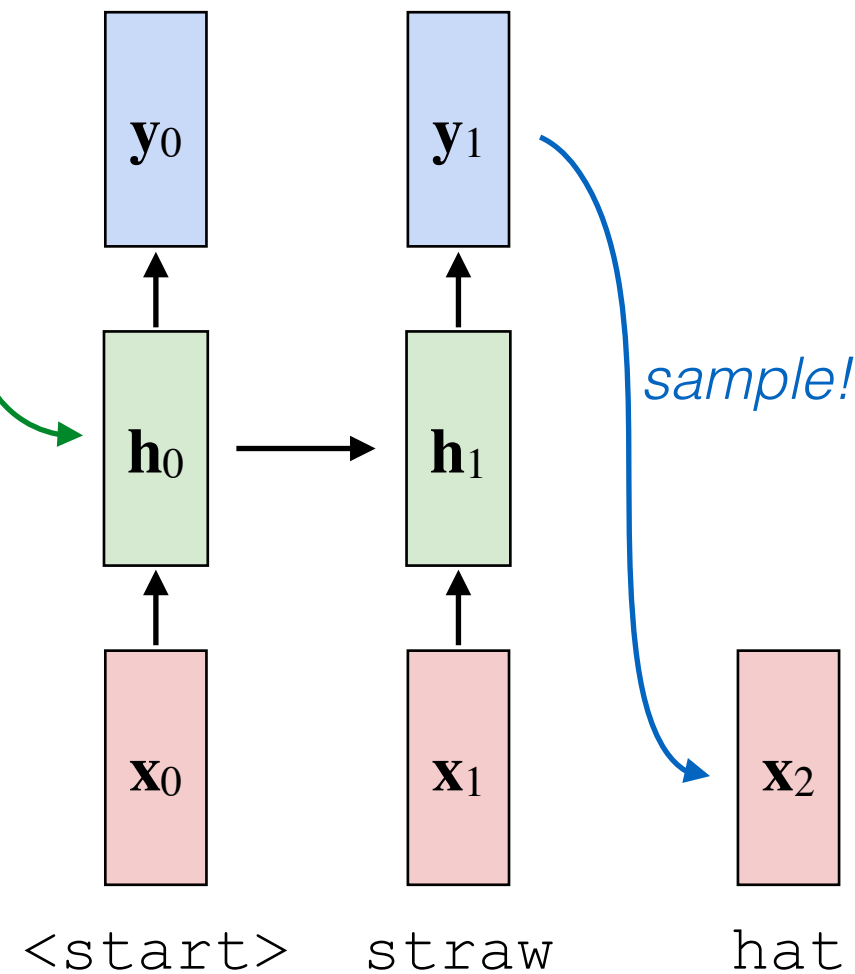
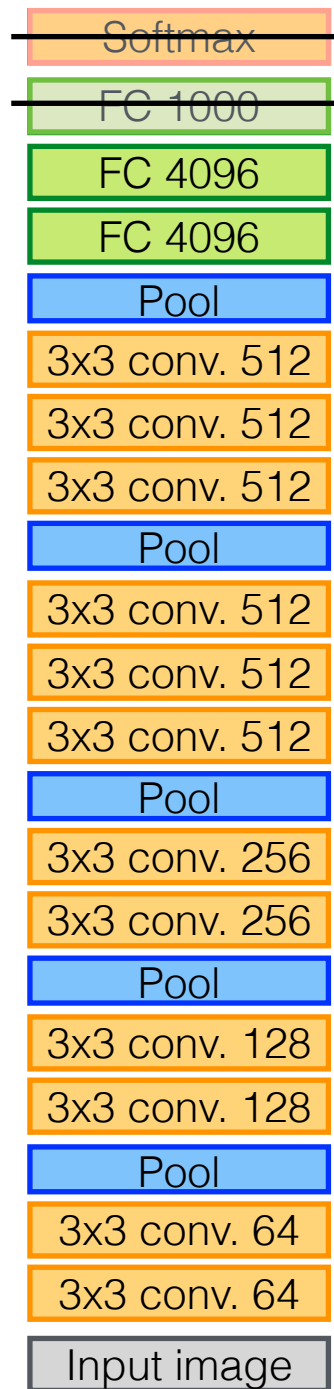


Image Captioning

Test Image



Pre-trained on
ImageNet



h is conditioned by the
image representation

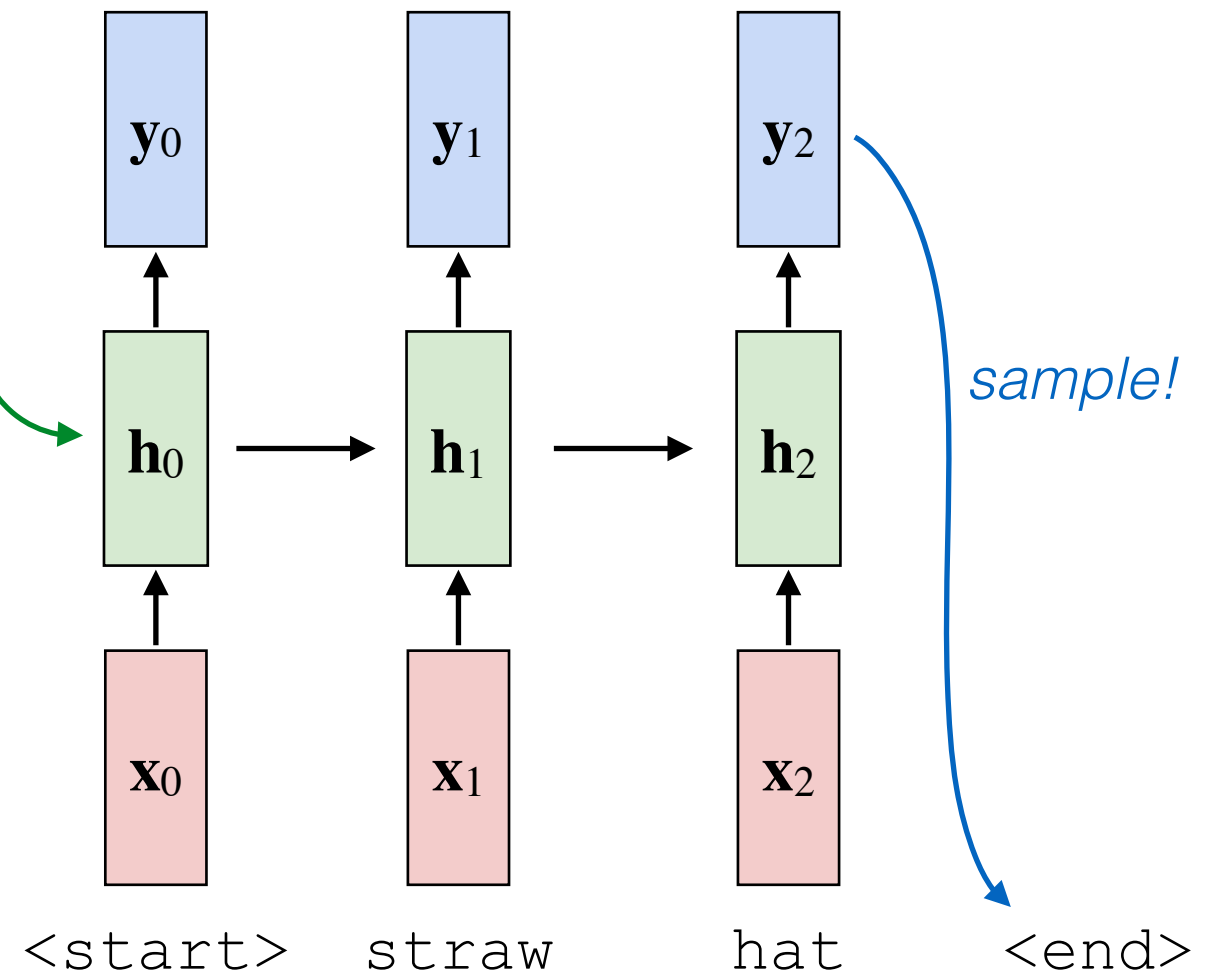


Image Captioning: results

(captions generated with neuraltalk2: <https://github.com/karpathy/neuraltalk2>)



A cat sitting on a suitcase on the floor



A cat is sitting on a tree branch



A dog is running in the grass with a frisbee



Two people walking on the beach with surfboards



A tennis player in action on the court



Two giraffes standing in a grassy field

Image Captioning: failure cases

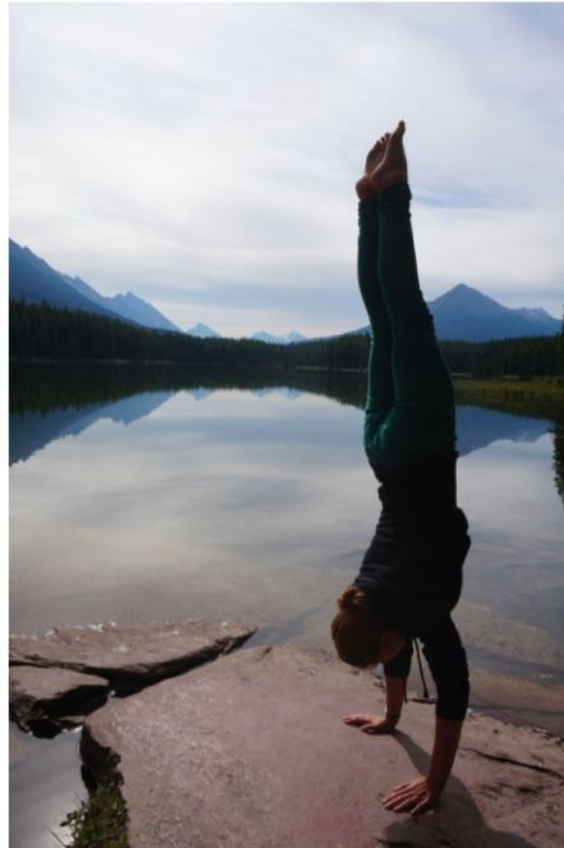
(captions generated with neuraltalk2: <https://github.com/karpathy/neuraltalk2>)



A woman is holding a cat in her hand



A person holding a computer mouse on a desk



A woman standing on a beach holding a surfboard

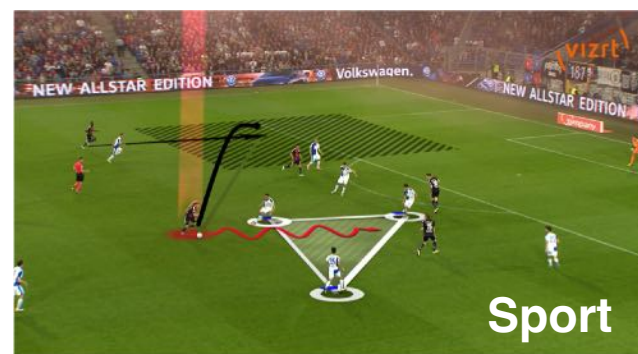
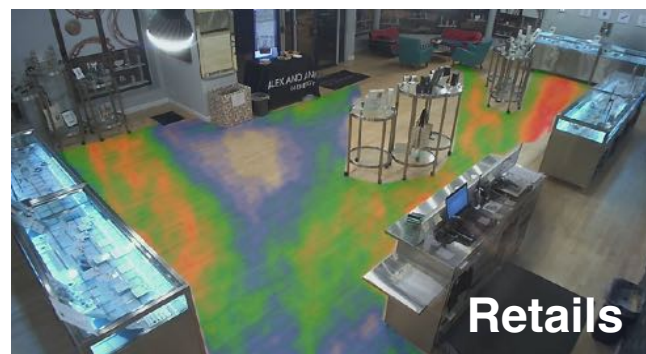


A bird is perched on a tree branch



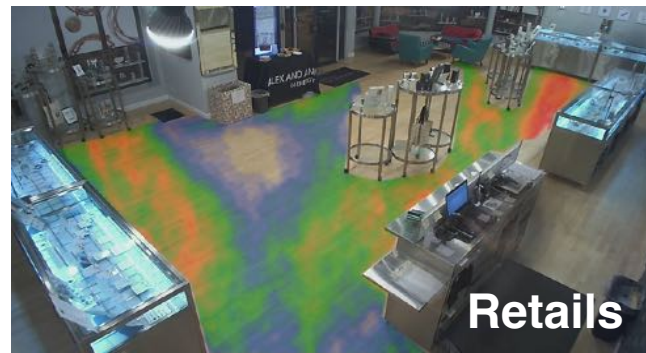
A man in a baseball uniform throwing a ball

A brief intro to predictive vision and motion prediction



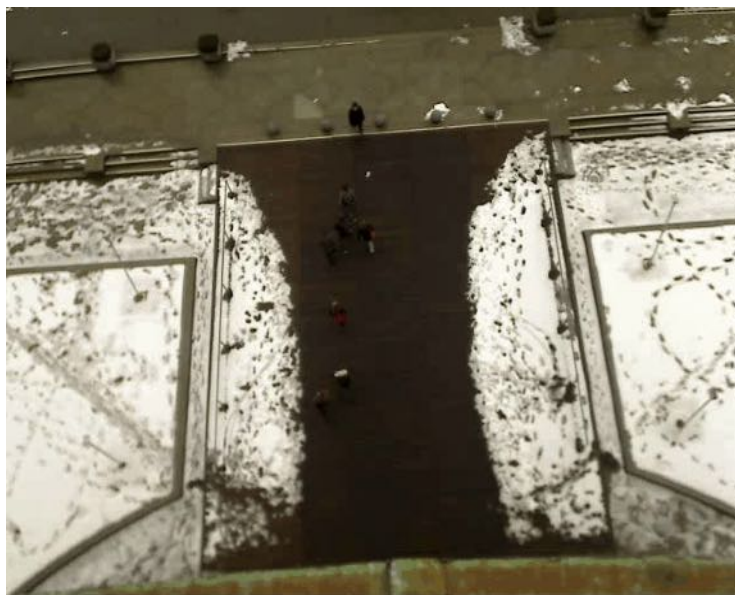
A brief intro to Predictive Vision

- Given a single picture and an observed “target”, humans are able to predict the most likely future



Humans in crowded spaces

- Humans motion in (a crowded) space is mostly influenced by the scene and the other active agents
 - E.g. Stanford Drone Dataset: videos of various agents that navigate a real world outdoor environment

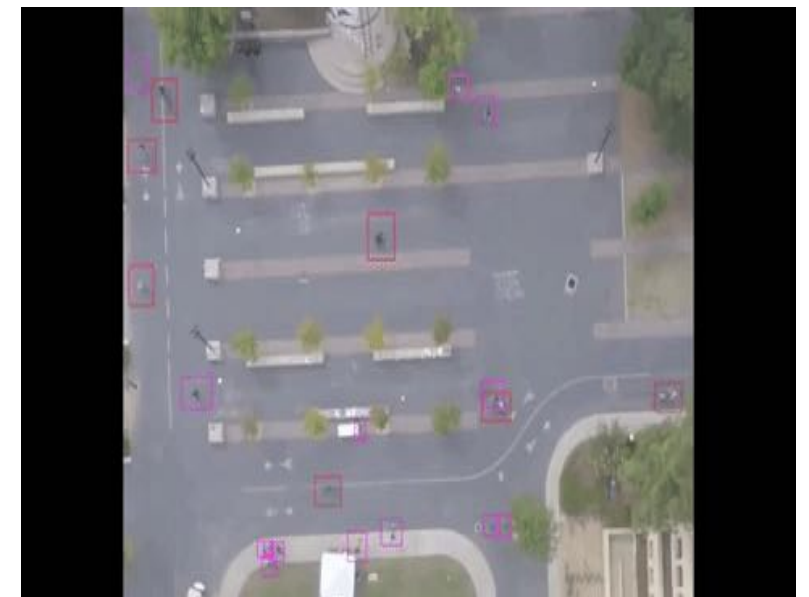


ETH
(ETH and HOTEL)



UCY
(ZARA1,ZARA2,UNIV)

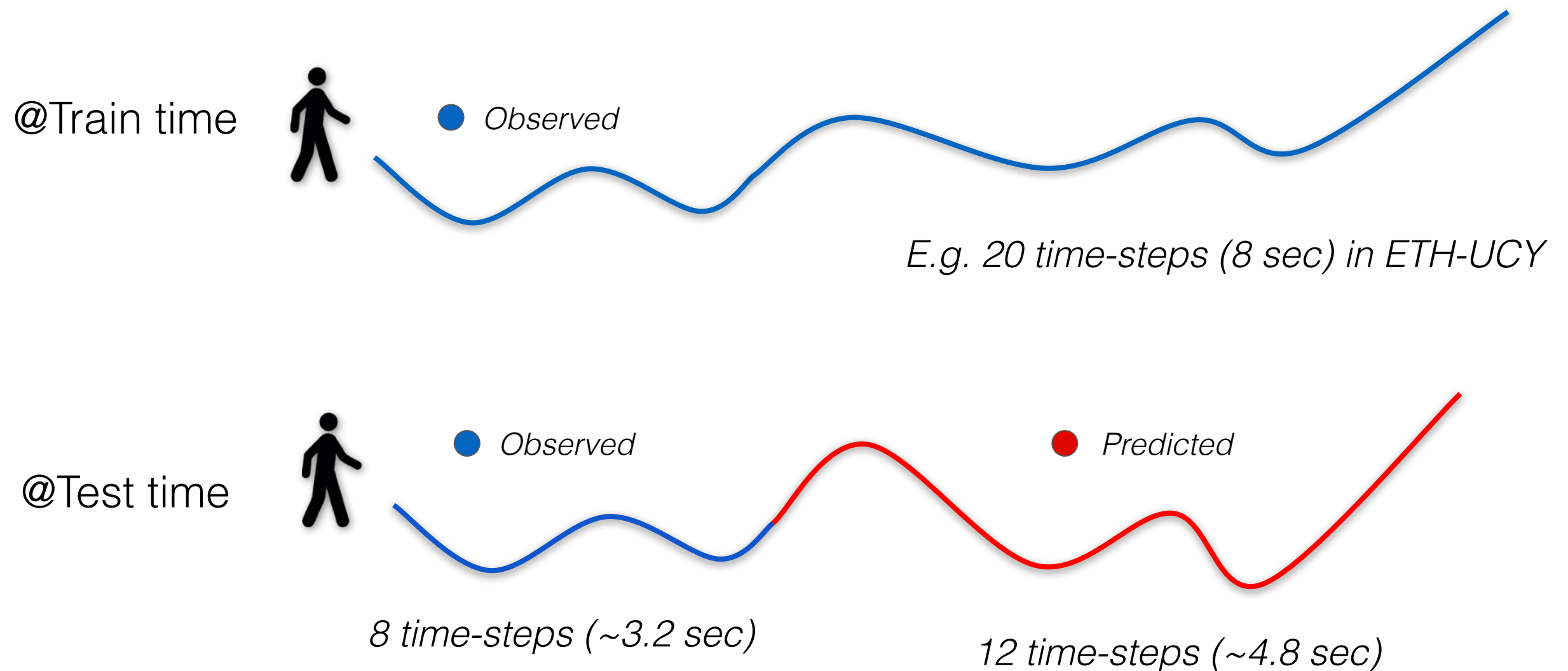
+



SDD

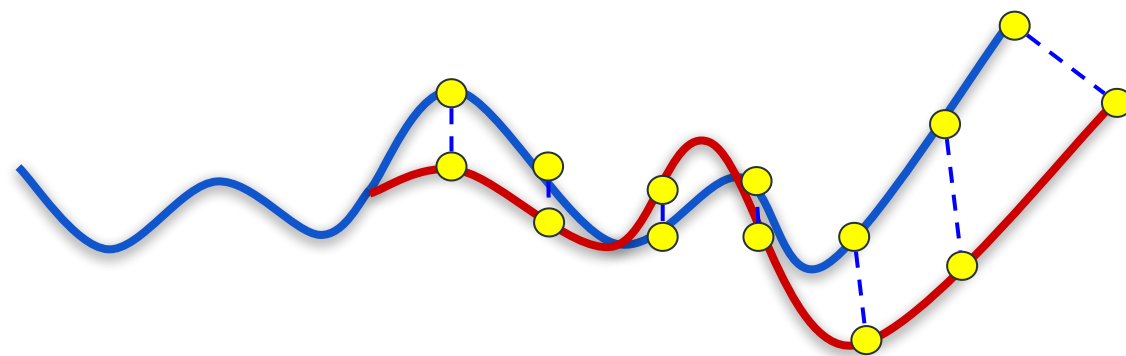
Goal: motion (*trajectory*) prediction

- Standard evolution protocol:
 - An agent is represented by its (x, y) coordinates

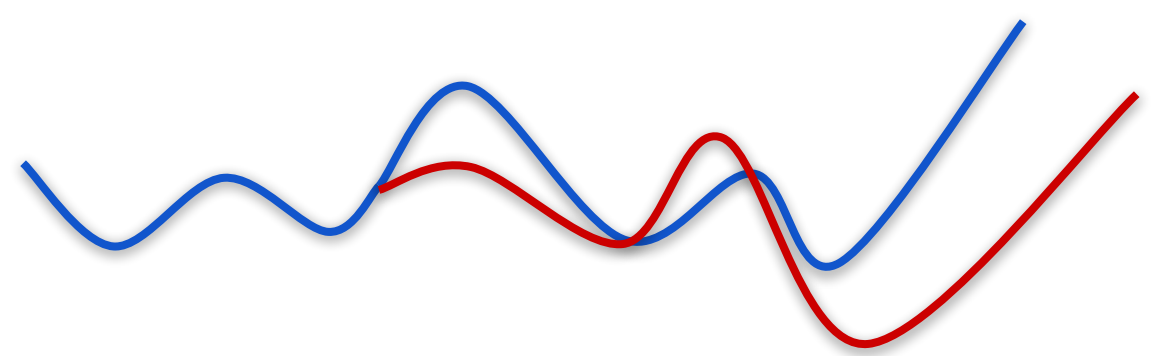


Goal: motion (*trajectory*) prediction

- Standard evolution protocol:
 - Given a predicted trajectory, we evaluate the quality of the prediction measuring the error w.r.t. the ground truth
 - Several Metrics: ADE, FDE (but also NLL)



Average Displacement Error
(ADE)

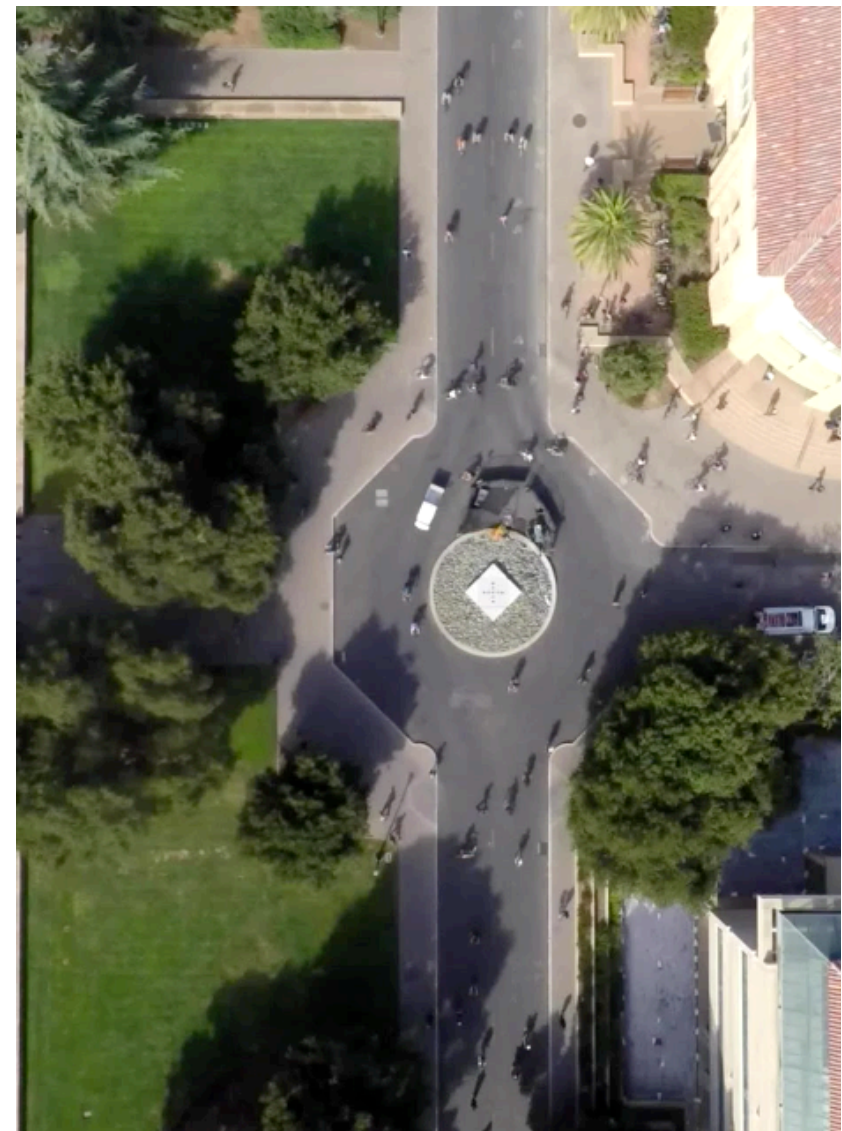


Final Displacement Error
(FDE)

Negative Log-Likelihood (NLL): it measures the fit of a ground-truth sample to the predicted distribution

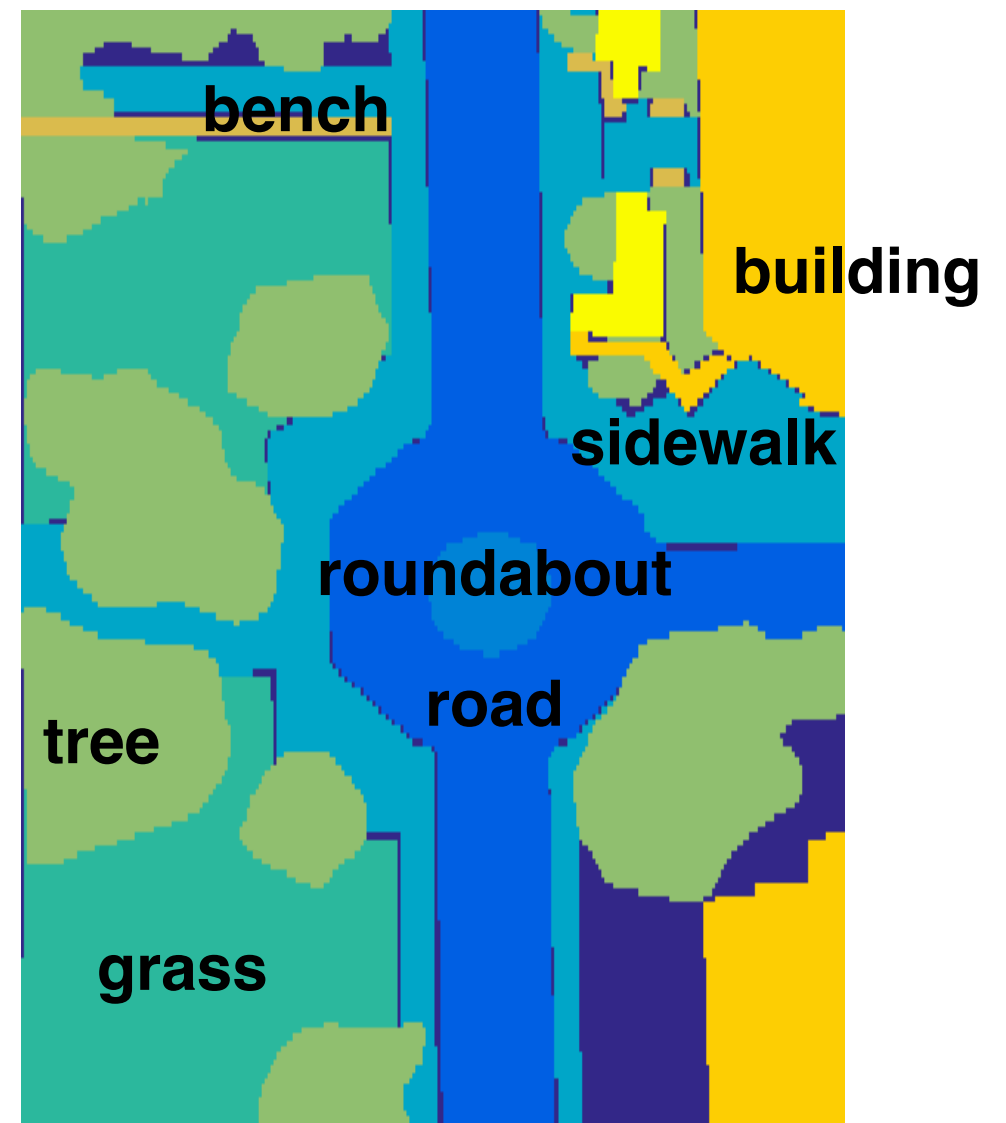
Motivation

- We believe this ability is mostly driven by two factors
 - the *dynamics* of active agents



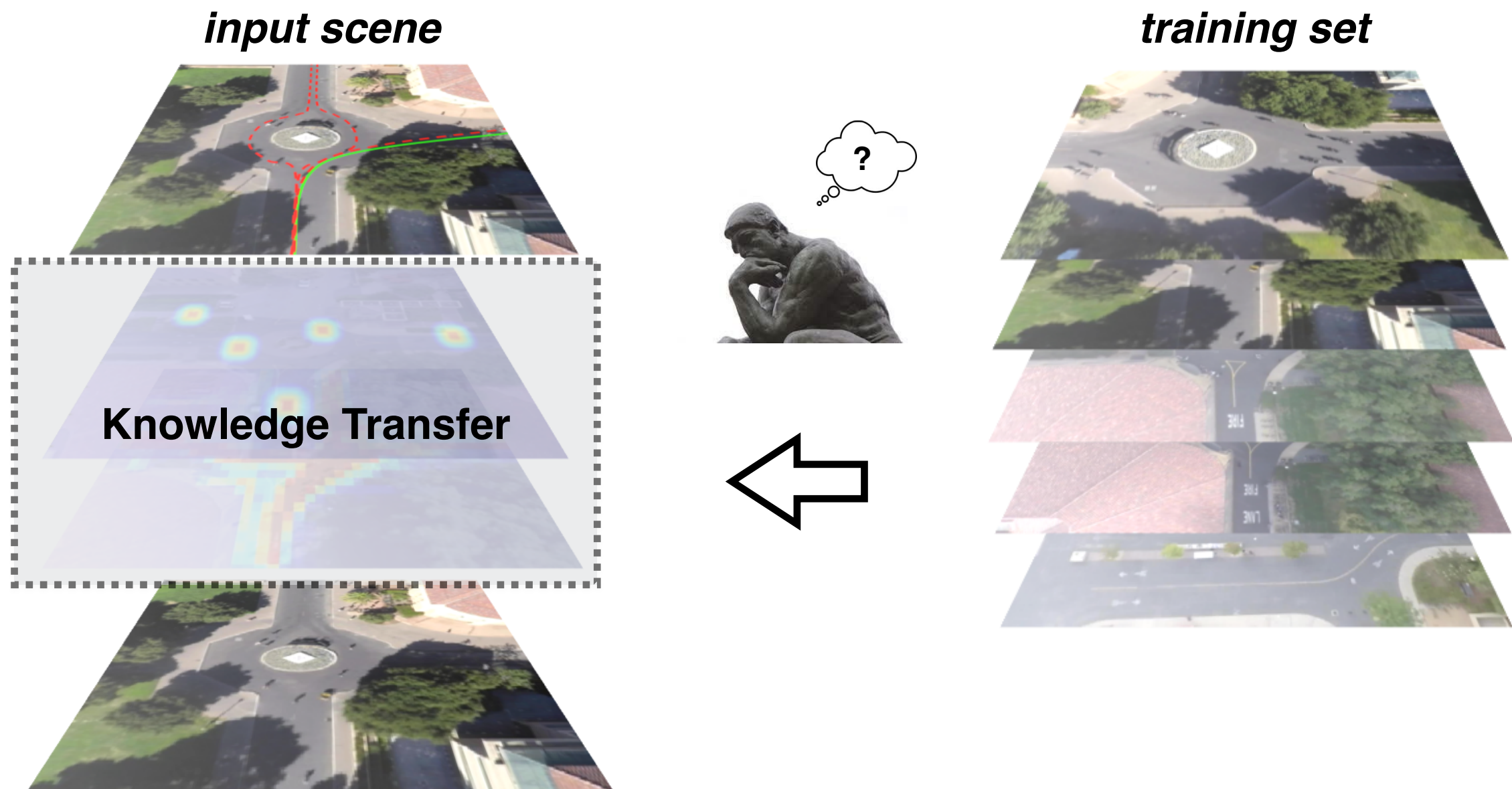
Motivation

- We believe this ability is mostly driven by two factors
 - the *dynamics* of active agents
 - the *semantic* of the scene



Approach

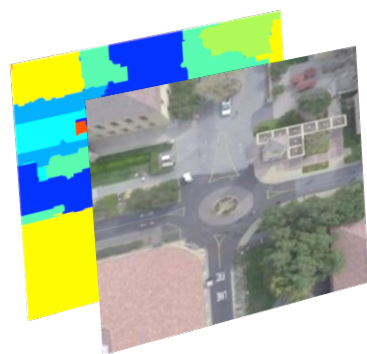
- This knowledge can be transferred to a new scene



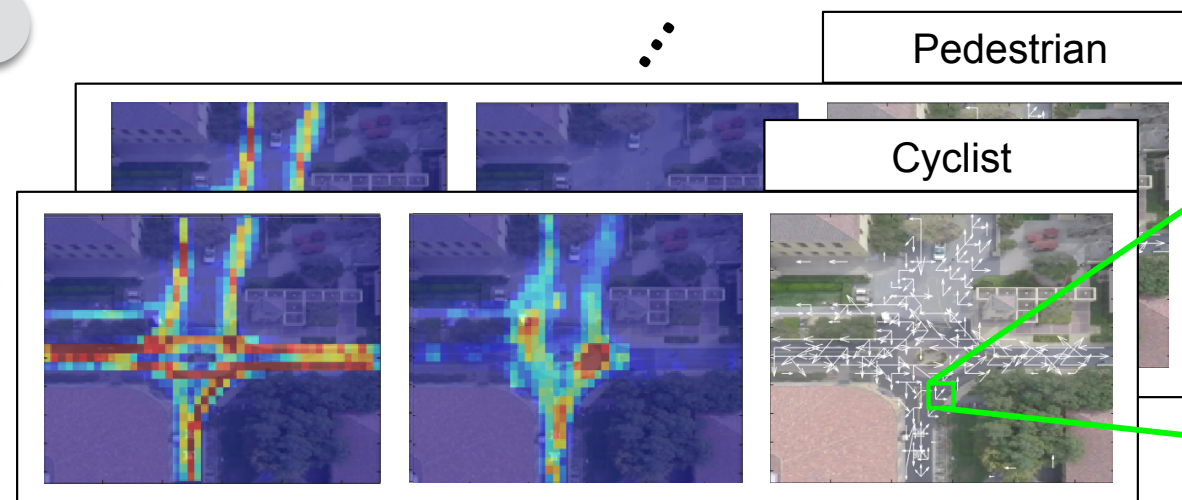
Approach

- Given an input scene we build a “*navigation map*” $\mathbf{M}$ which collects the navigation statistics
- In our early works, for each patch in $\mathbf{M}$ we collect:
 - Popularity score, Routing score, Histogram of Directions and Histogram of Speeds

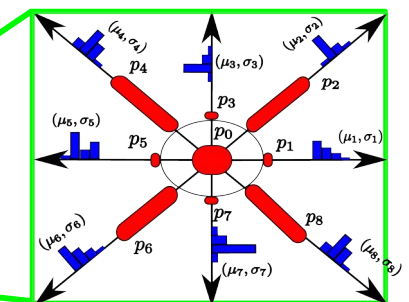
“Hand-crafted” features



(a) Input scene



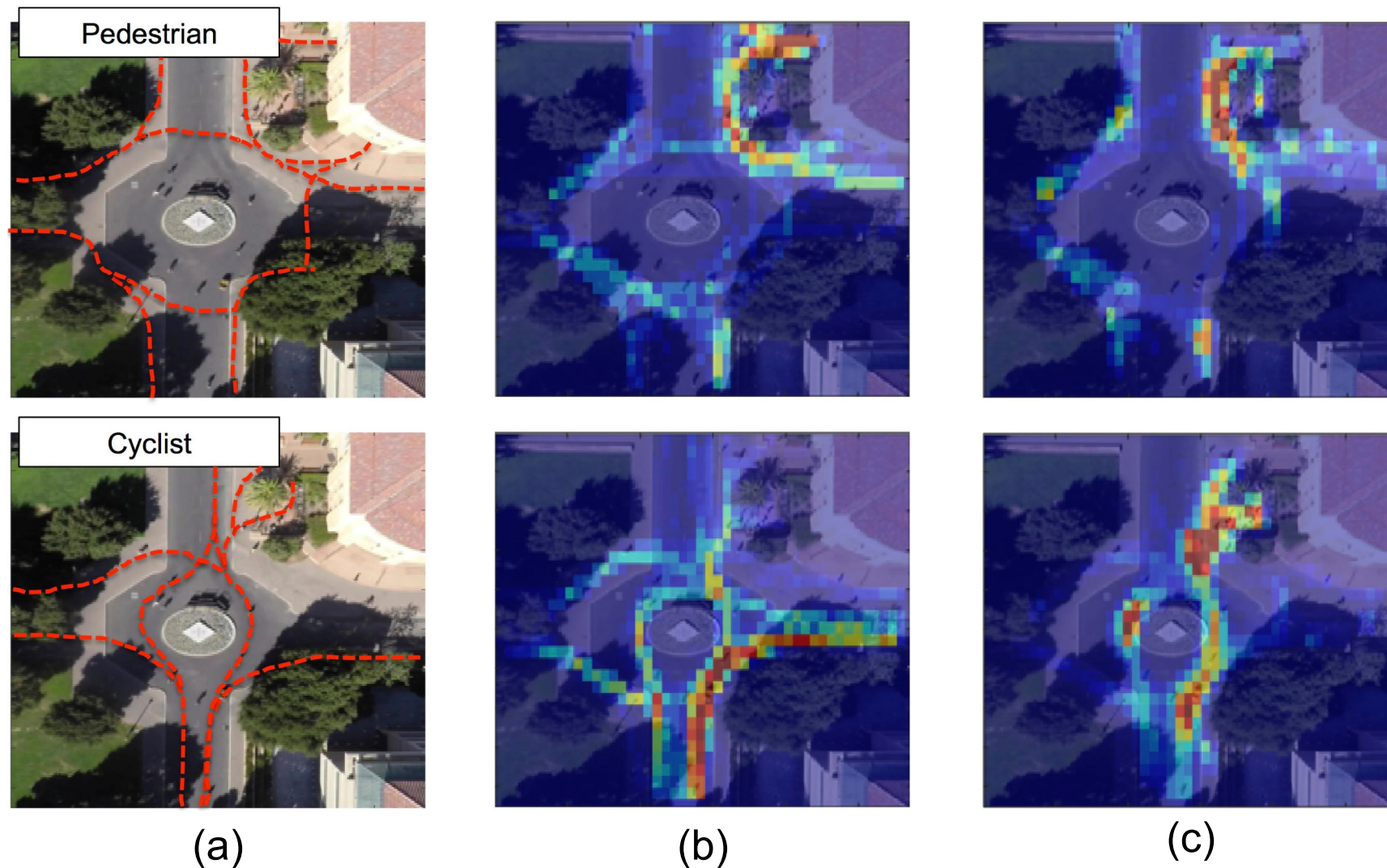
(b) “Navigation map”



(c) *HoD* and *HoS*

Prediction model

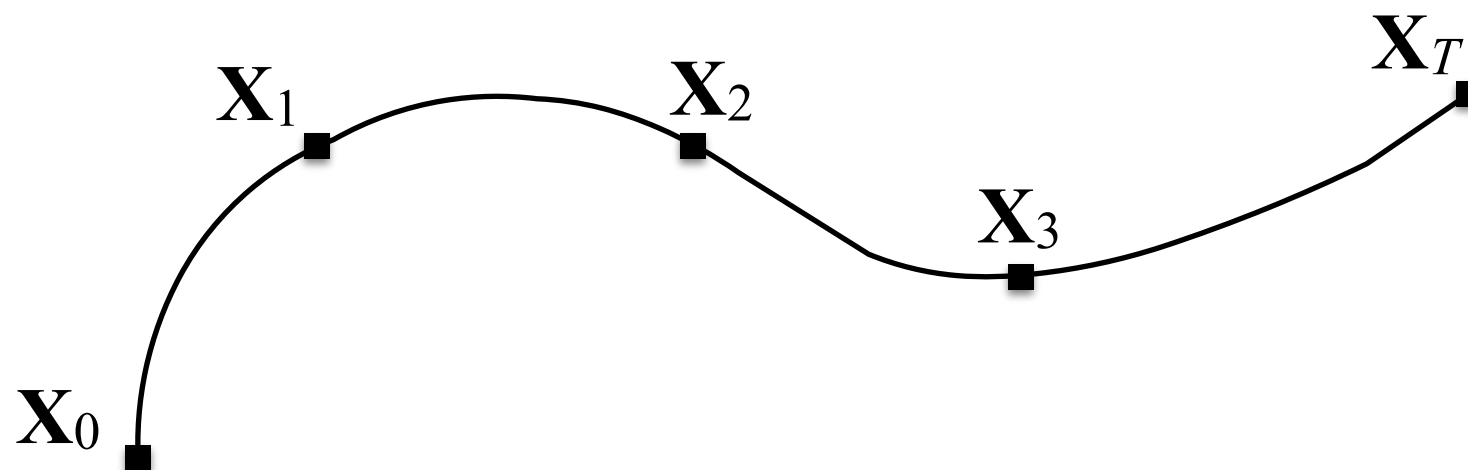
- Navigation Map: qualitative examples



Column (a) visualizes the most common paths for both classes.
Columns (b, c) show the corresponding popularity and routing maps.

Prediction model

- The target state variable is defined as $\mathbf{X}_k = (\mathbf{P}_k, \mathbf{V}_k)^T$
 - $\mathbf{P}_k = (X_k, Y_k)^T$ (position) and $\mathbf{V}_k = (\Omega_k, \Theta_k)^T$ (velocity)
- The target interacts with the map $\mathbf{M}$ by exploiting the navigation values for the patch he is occupying
- Given an initial state $\mathbf{X}_0$, our goal is to generate a sequence of future states $\mathbf{X}_1, \dots, \mathbf{X}_T$ (*i.e.*, a trajectory)



Prediction model

- The dynamic process describing the target motion is defined by:
 - $\mathbf{P}_{k+1} = \mathbf{P}_k + (\Omega_k \cos \Theta_k, \Omega_k \sin \Theta_k)' + \mathbf{w}_k$ (constant velocity)
 - $\mathbf{V}_{k+1} = \Phi(\mathbf{P}_k, \mathbf{V}_k; \mathbf{M})$
- The learned expected values in $\mathbf{M}$ allows our model to generate non-linear behaviors
- $\Phi(\cdot)$ is defined in probabilistic terms by means of a Dynamic Bayesian Network (DBN)

See also: [Coscia, Castaldo, Palmieri, Alahi, Savarese, **Ballan** - IVC'18]

(focus on long-term trajectory prediction)

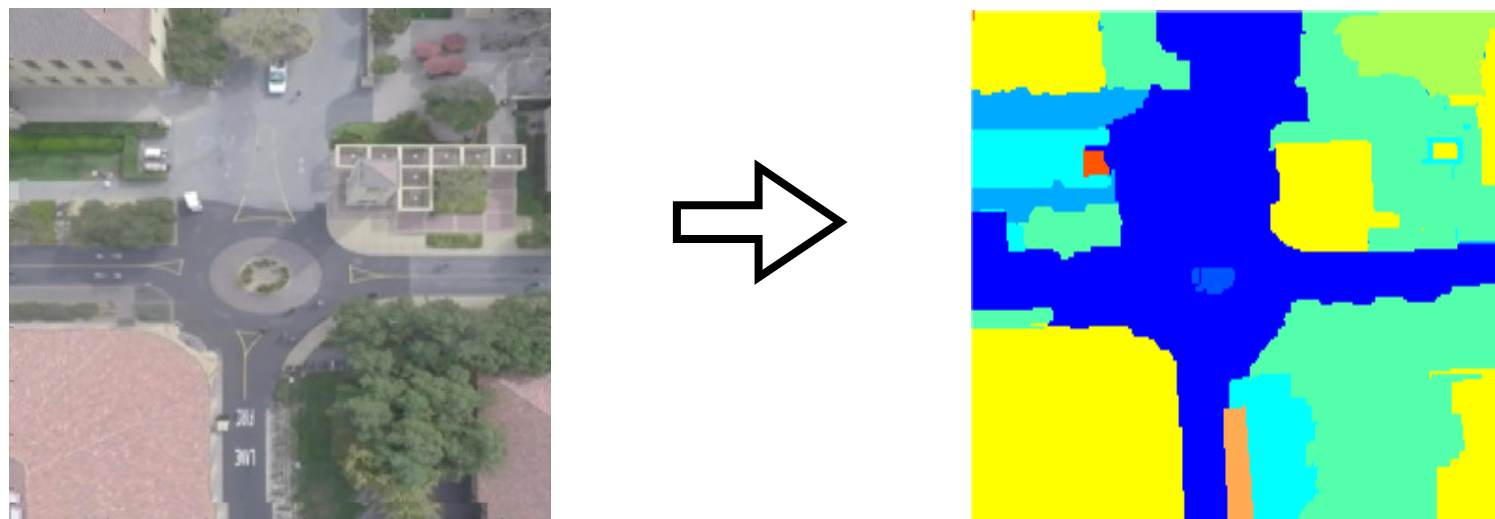
IVC Best Paper Award 2021



Knowledge transfer

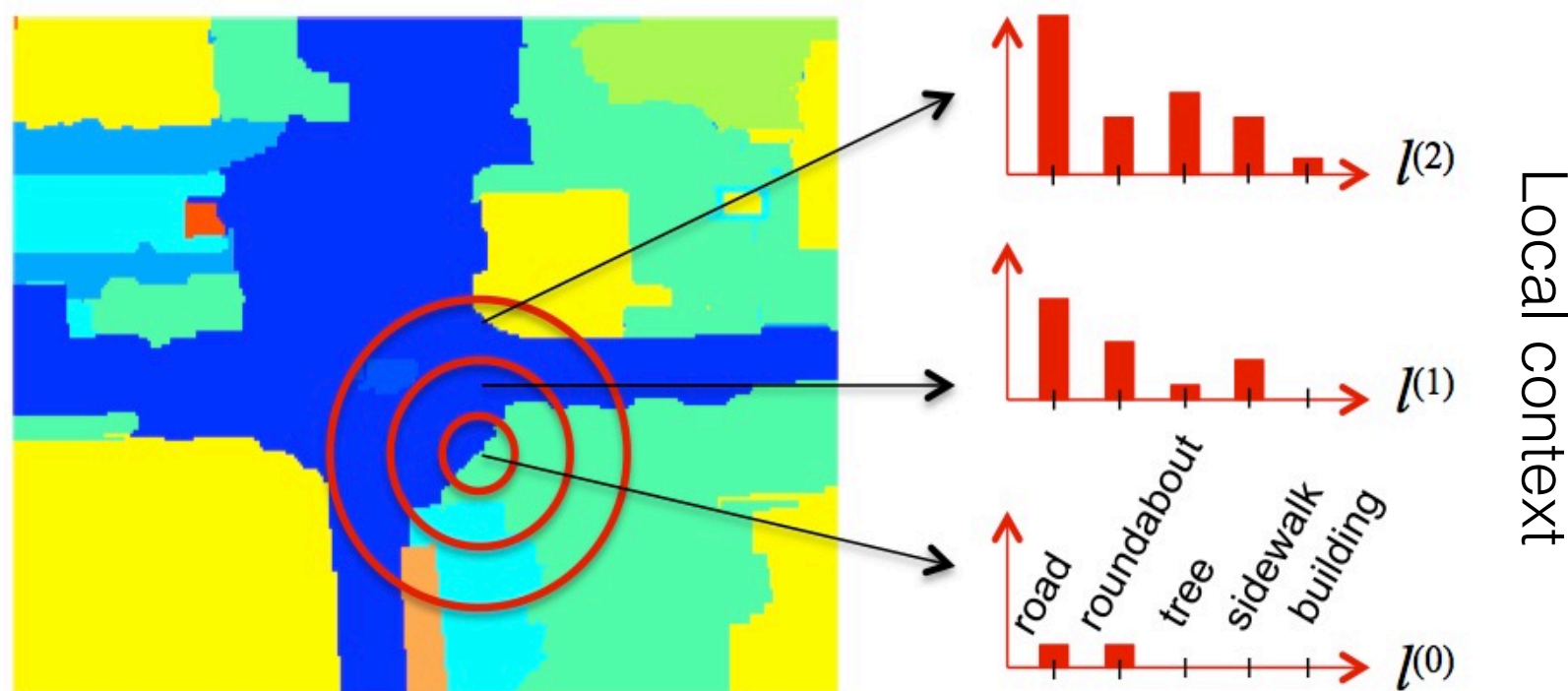
“The elements of the scene define a semantic context, and they might determine similar behaviors in scenes characterized by a similar context”

- Our data-driven approach uses scene similarity to transfer the functional properties to a new scene
- Scene parsing: we use a “non-parametric” algorithm (based on local features SIFT+LLC, GIST and MRF inference)



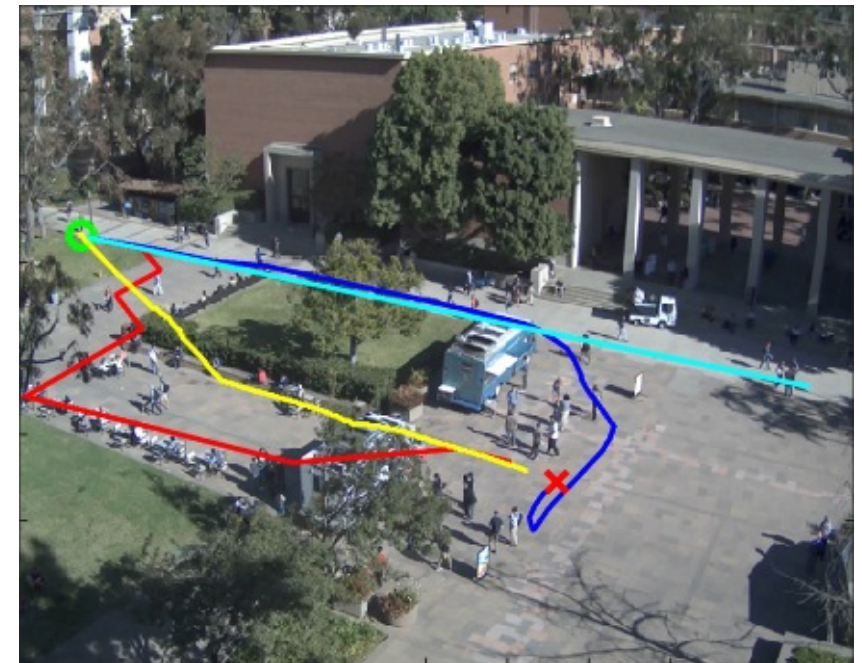
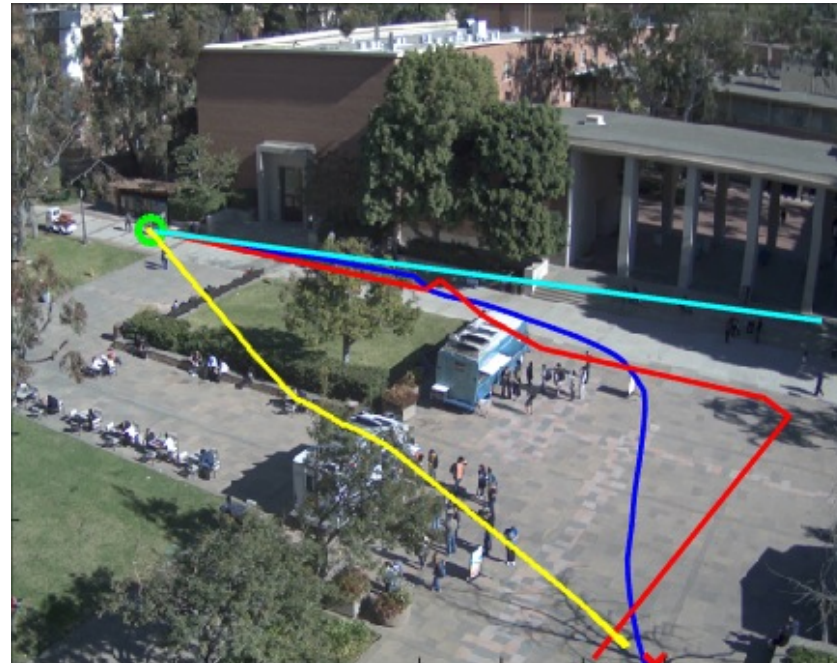
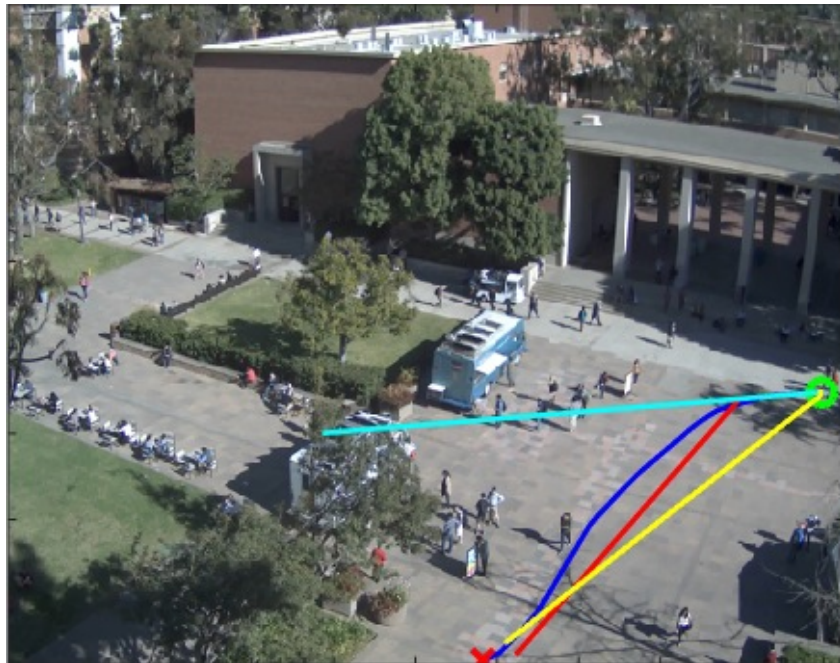
Knowledge transfer

- Context Descriptors: a weighted concatenation of the *global* and *local* semantic context components
 - *global context*: vector of distances between classes
 - *local context*: encodes the spatial configuration of nearby patches at multiple levels



Results: path prediction

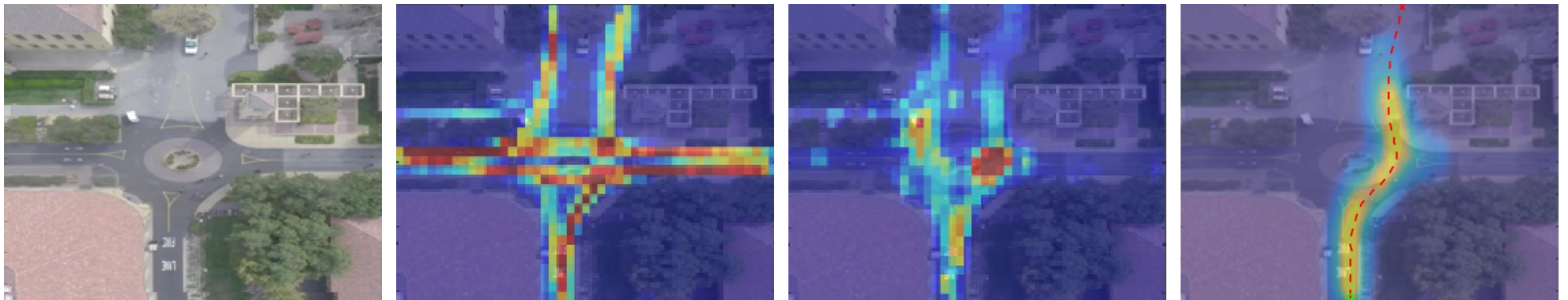
- Qualitative examples on the UCLA-courtyard scene (*blue* is ground-truth, *cyan* is LP, *yellow* is IOC, *red* is ours)



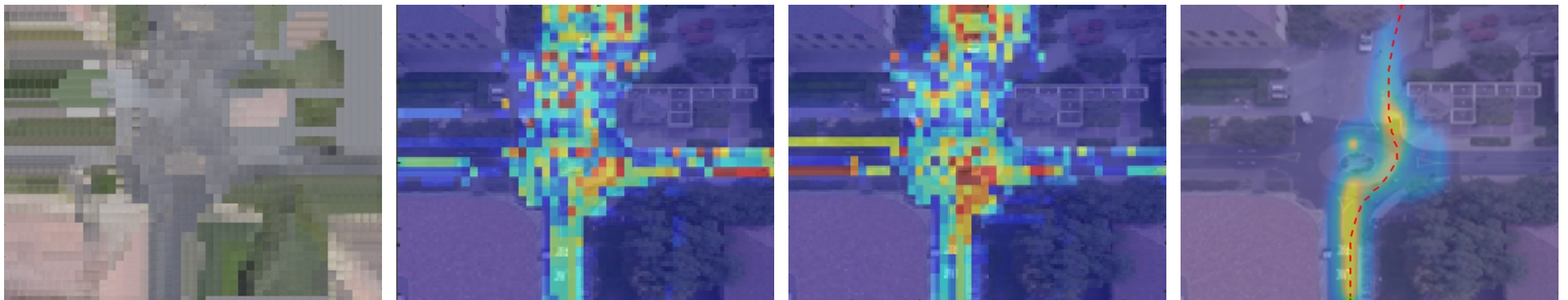
Results: knowledge transfer

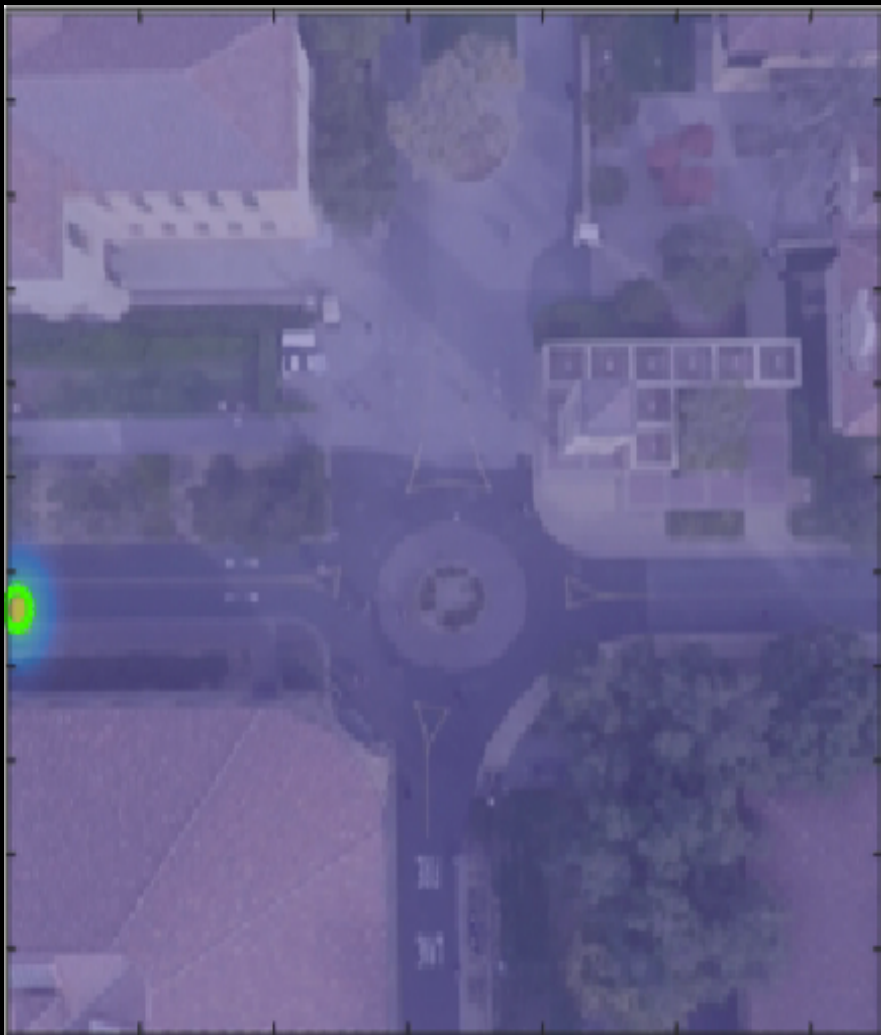
- Qualitative examples:
 - (1) path forecasting vs (2) knowledge transfer

(1)



(2)





starting point



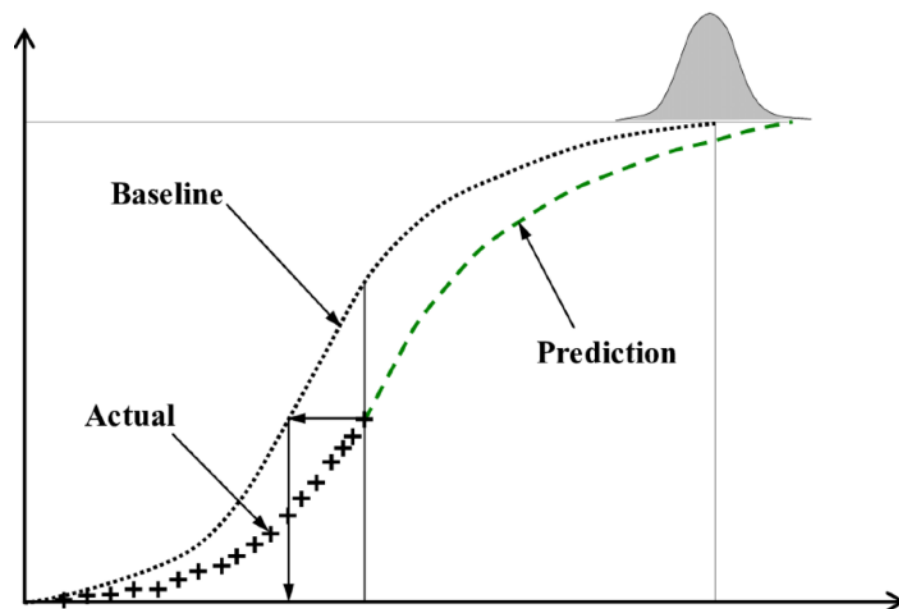
heatmap



path prediction

Trajectory prediction

- A broad range of approaches:



*Based on Bayesian models /
Kalman-based Filtering*

“Traditional approaches”

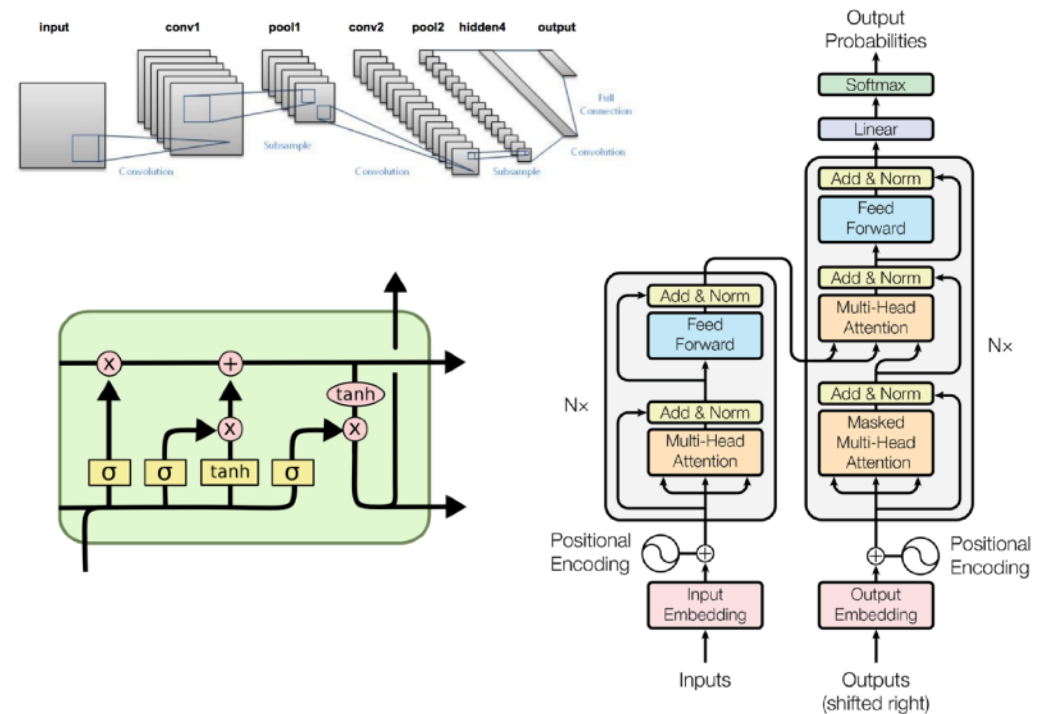


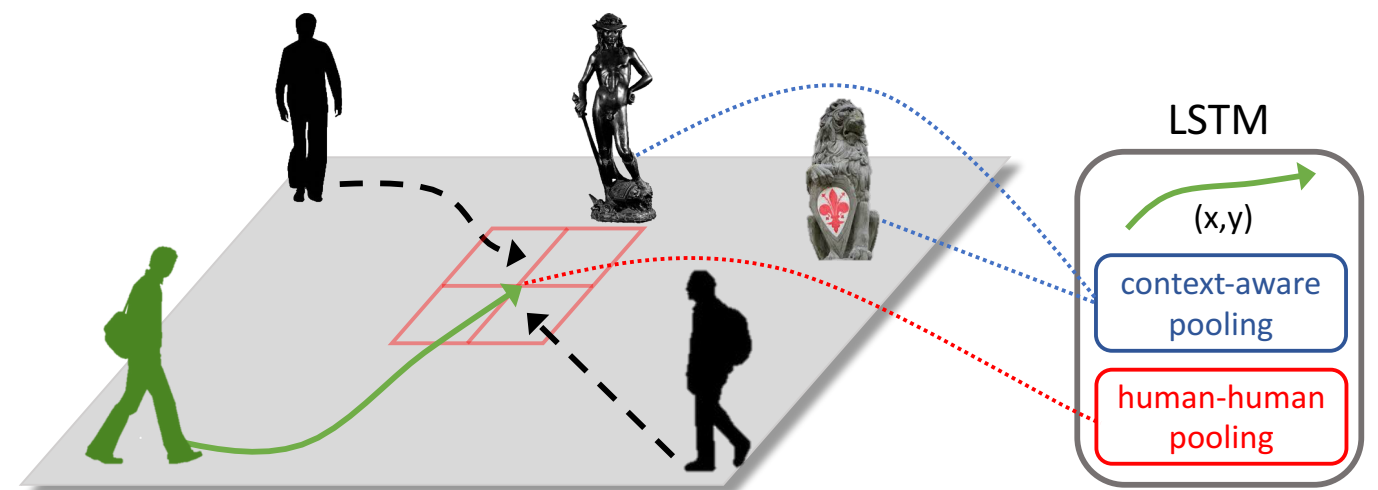
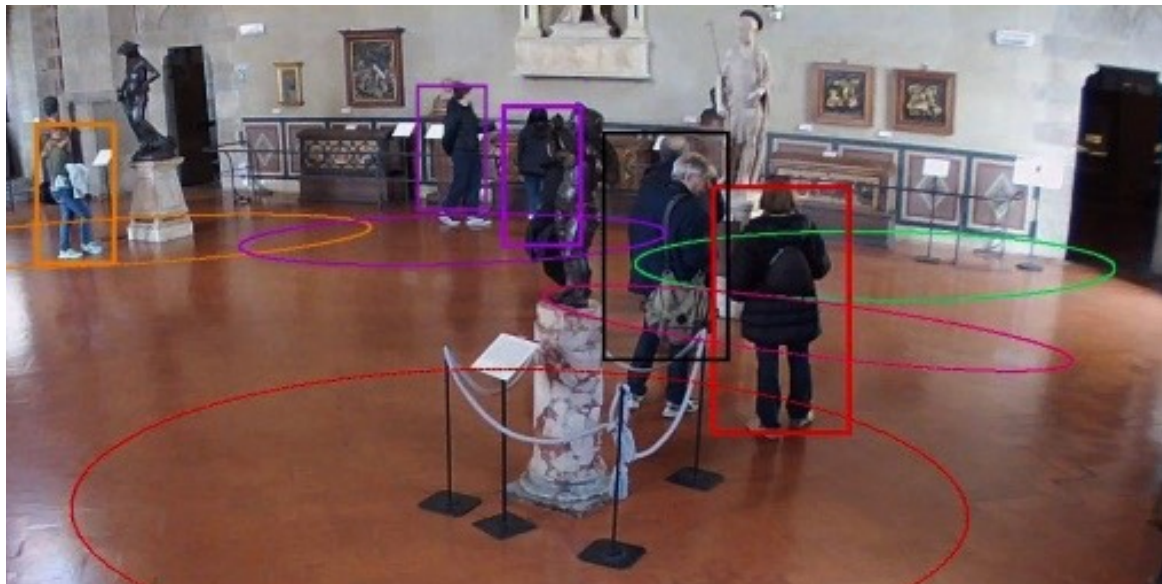
Figure 1: The Transformer - model architecture.

Based on Deep-NN Architectures

Current paradigm

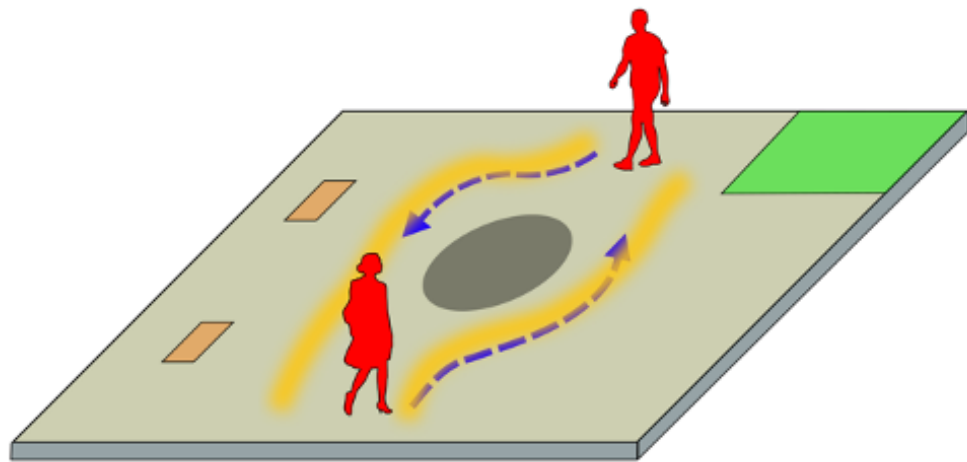
Context-aware LSTM

- Trajectories are modelled using RNNs/LSTMs
- A context-aware pooling layer combining:
 - Human-human interactions
 - Human-space interactions



Context-aware LSTM

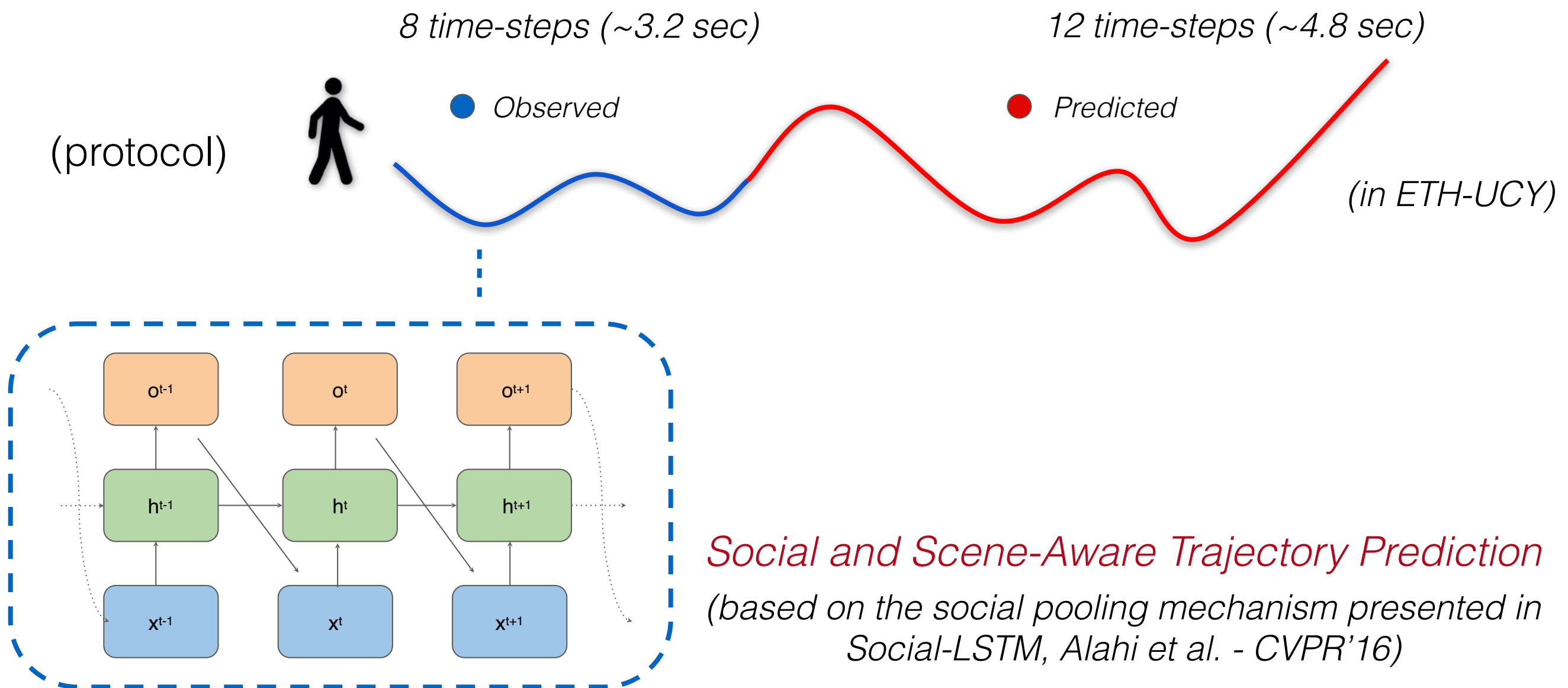
- Trajectories are modelled using RNNs/LSTMs
- A context-aware pooling layer combining:
 - Human-human interactions (*social* pooling)
(+ *navigation* pooling)
 - Human-space interactions (*semantic* pooling)



Social and Scene-Aware Trajectory Prediction
(based on the social pooling mechanism presented in
Social-LSTM, Alahi et al. - CVPR'16)

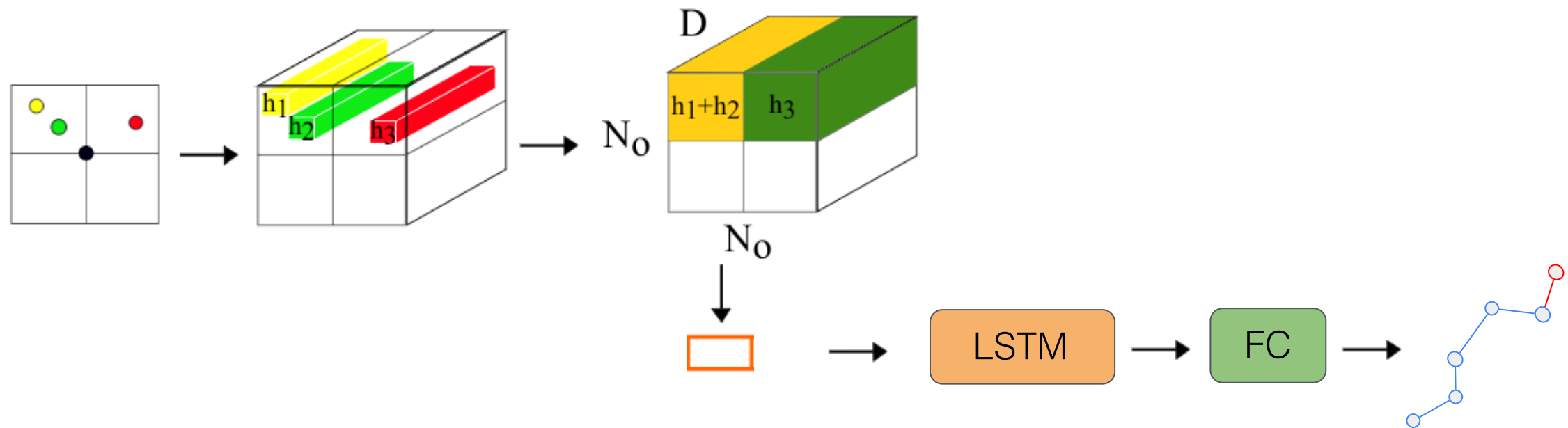
Context-aware LSTM

- Trajectories are modelled using RNNs/LSTMs



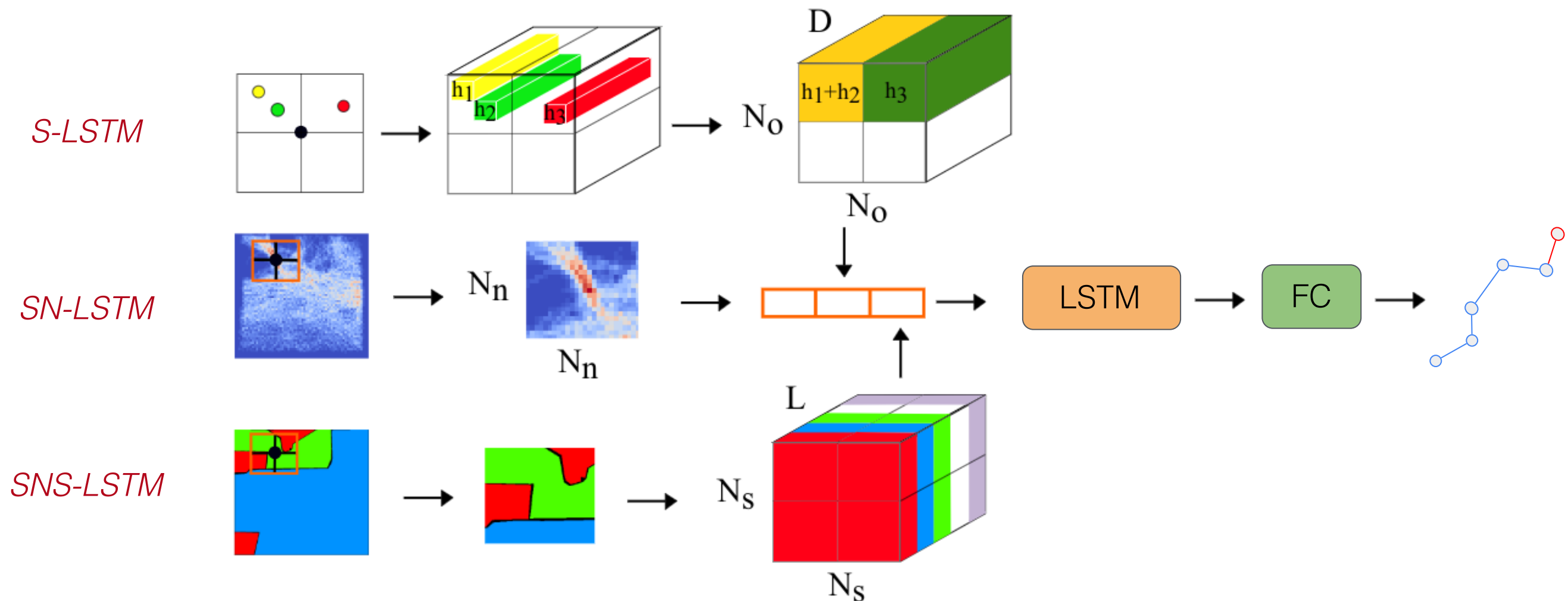
Context-aware LSTM

- Social and scene-aware trajectory prediction:



Context-aware LSTM

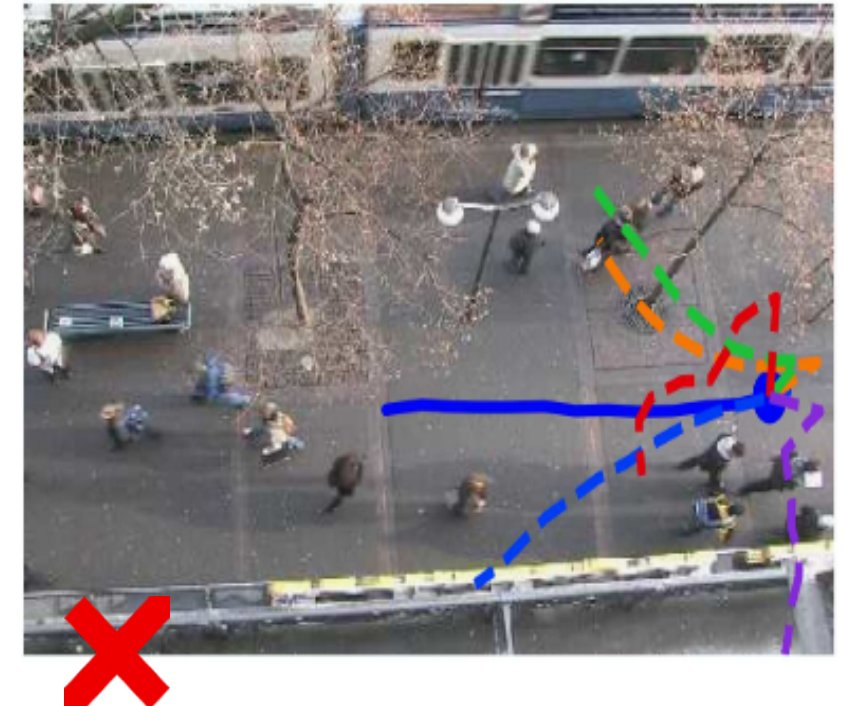
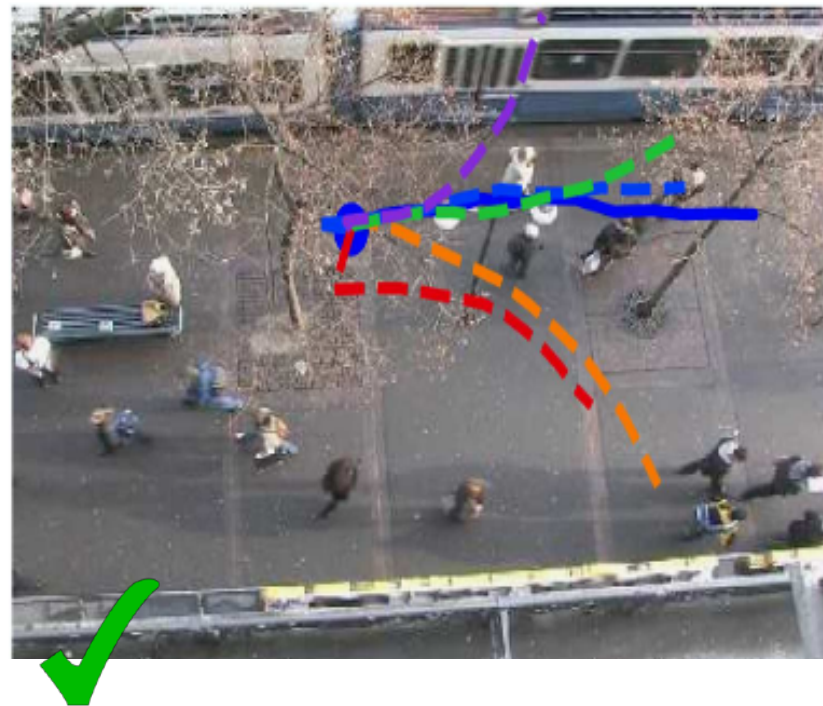
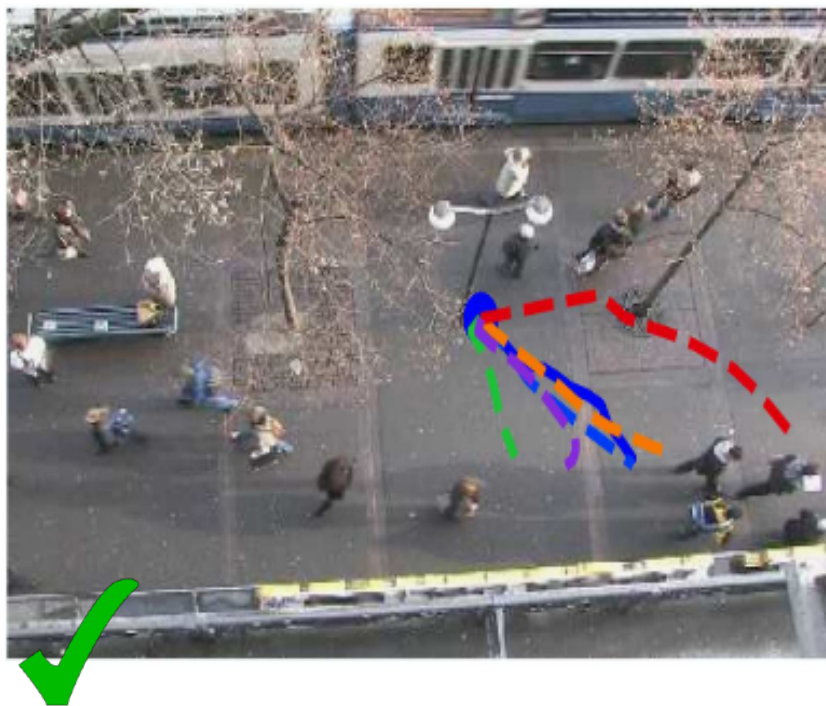
- Social and scene-aware trajectory prediction:



Experiments

- Qualitative results on ETH/UCY:

Ground-truth SS-LSTM S-LSTM
SNS-LSTM SN-LSTM Vanilla LSTM



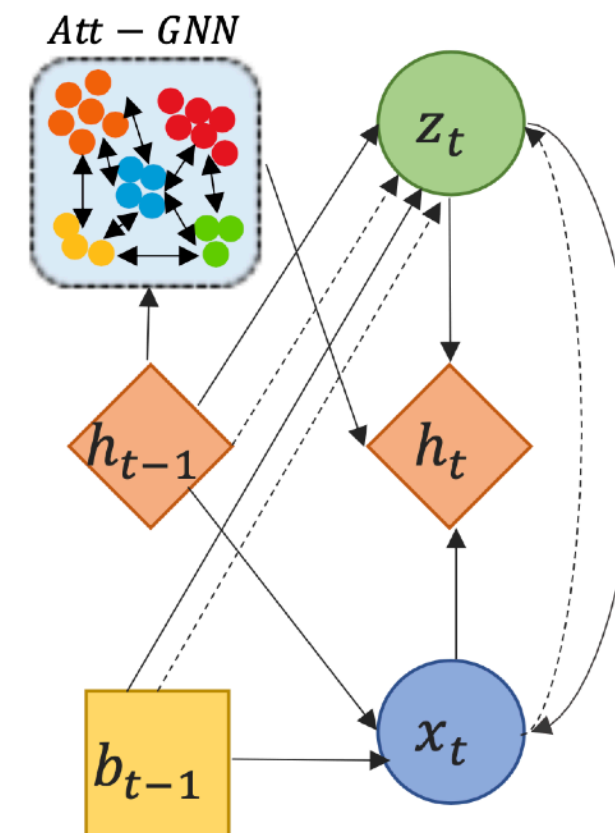
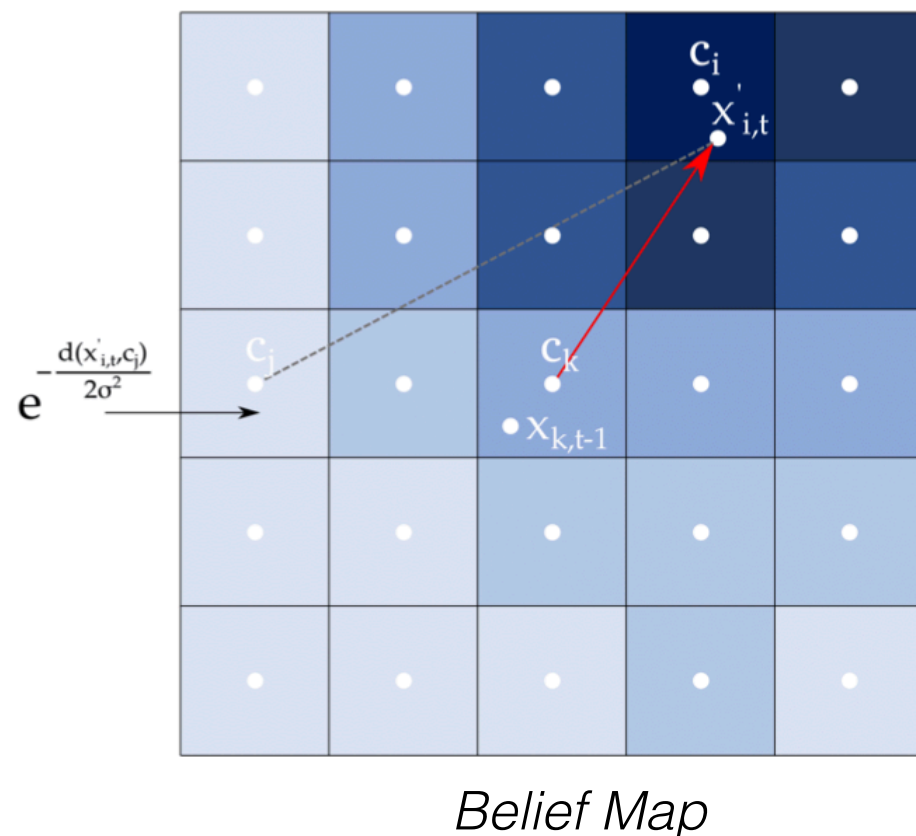
Experiments

- Quantitative results on ETH/UCY:
 - Baseline (backbone): “Vanilla” LSTM

(ADE/FDE)	ETH	HOTEL	UNIV	ZARA1	ZARA2	AVG
“Vanilla” LSTM	0.52/2.84	0.33/1.90	0.52/2.92	0.41/2.35	0.27/1.48	0.41/2.30
S-LSTM	0.51/2.82	0.31/1.67	0.55/3.04	0.36/2.05	0.25/1.42	0.40/2.20
SN-LSTM (ours)	0.47/2.55	0.44/2.25	0.39/2.10	0.29/1.56	0.28/1.59	0.37/2.01
SS-LSTM (ours)	0.48/2.57	0.24/1.38	0.43/2.54	0.33/1.81	0.31/1.63	0.36/1.99
SNS-LSTM (ours)	0.58/ 2.43	0.30/1.58	0.37/2.08	0.28/1.53	0.26/1.44	0.36/1.81

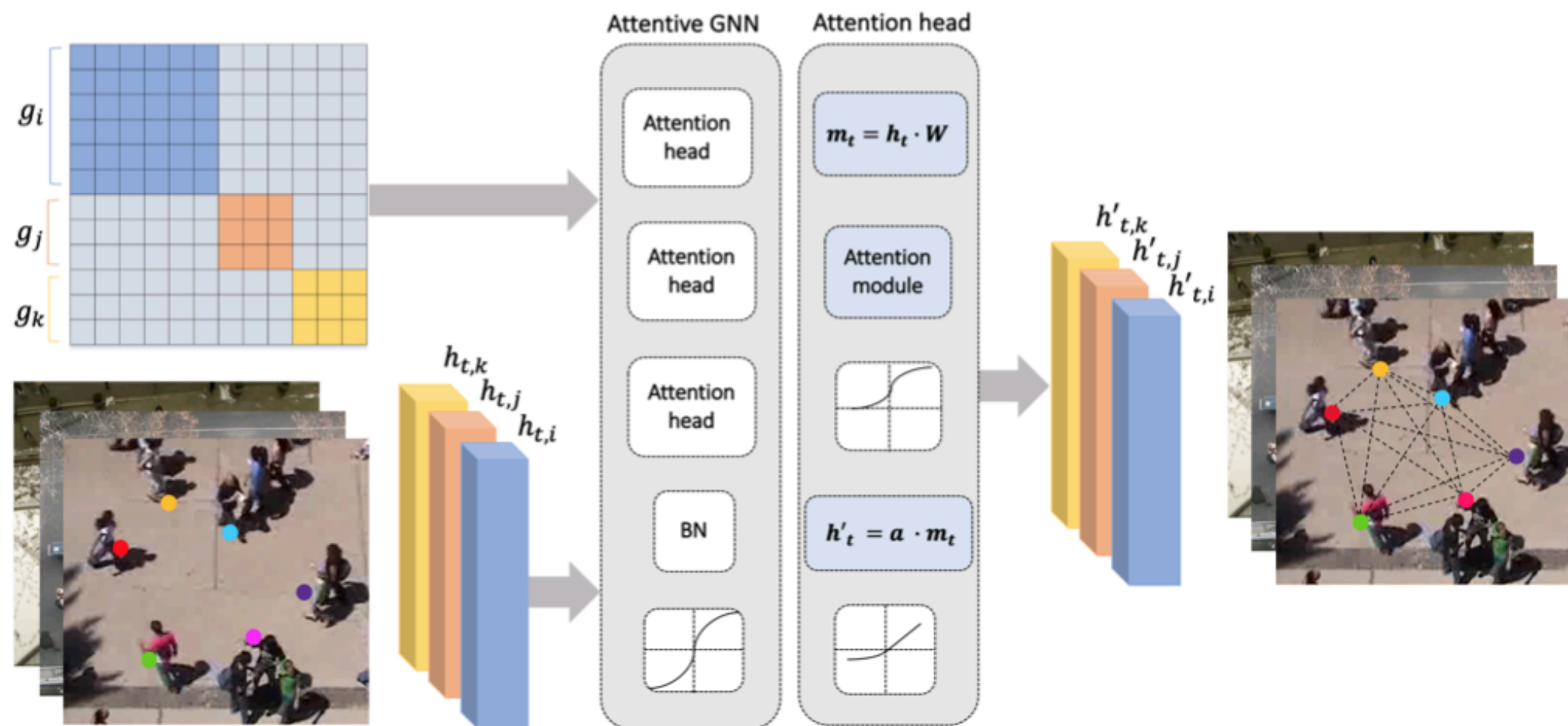
Attentive Conditional-VRNN

- A recurrent variational autoencoder is conditioned on prior belief maps
 - The RNN hidden state is refined with an attention module
 - At inference time, it generates future displacements



Attentive Conditional-VRNN

- Our model handles human interactions using an attentive hidden state refinement of our RNN
 - It's done by a graph NN based on an attention mechanism to learn relative weights between connected nodes



Experiments

- Quantitative results on ETH/UCY:

(ADE/FDE)	ETH	HOTEL	UNIV	ZARA1	ZARA2	AVG
S-LSTM CVPR'16	1.09/2.35	0.79/1.76	0.67/1.40	0.47/1.00	0.56/1.17	0.72/1.54
S-GAN CVPR'18	0.81/1.52	0.72/1.61	0.60/1.26	0.34/0.69	0.42/0.84	0.58/1.18
Trajectron ICCV'19	0.59/1.14	0.35/0.66	0.54/1.13	0.43/0.83	0.43/0.85	0.56/1.14
SoPhie CVPR'19	0.70/1.43	0.76/1.67	0.30/0.63	0.38/0.78	0.54/1.24	0.54/1.15
STGAT ICCV'19	0.78/1.60	0.30/0.54	0.51/1.08	0.33/0.72	0.29/0.63	0.44/0.91
AC-VRNN (ours)	0.61/ 1.09	0.30/0.55	0.58/1.22	0.34/ 0.68	0.28/ 0.59	0.42/0.83

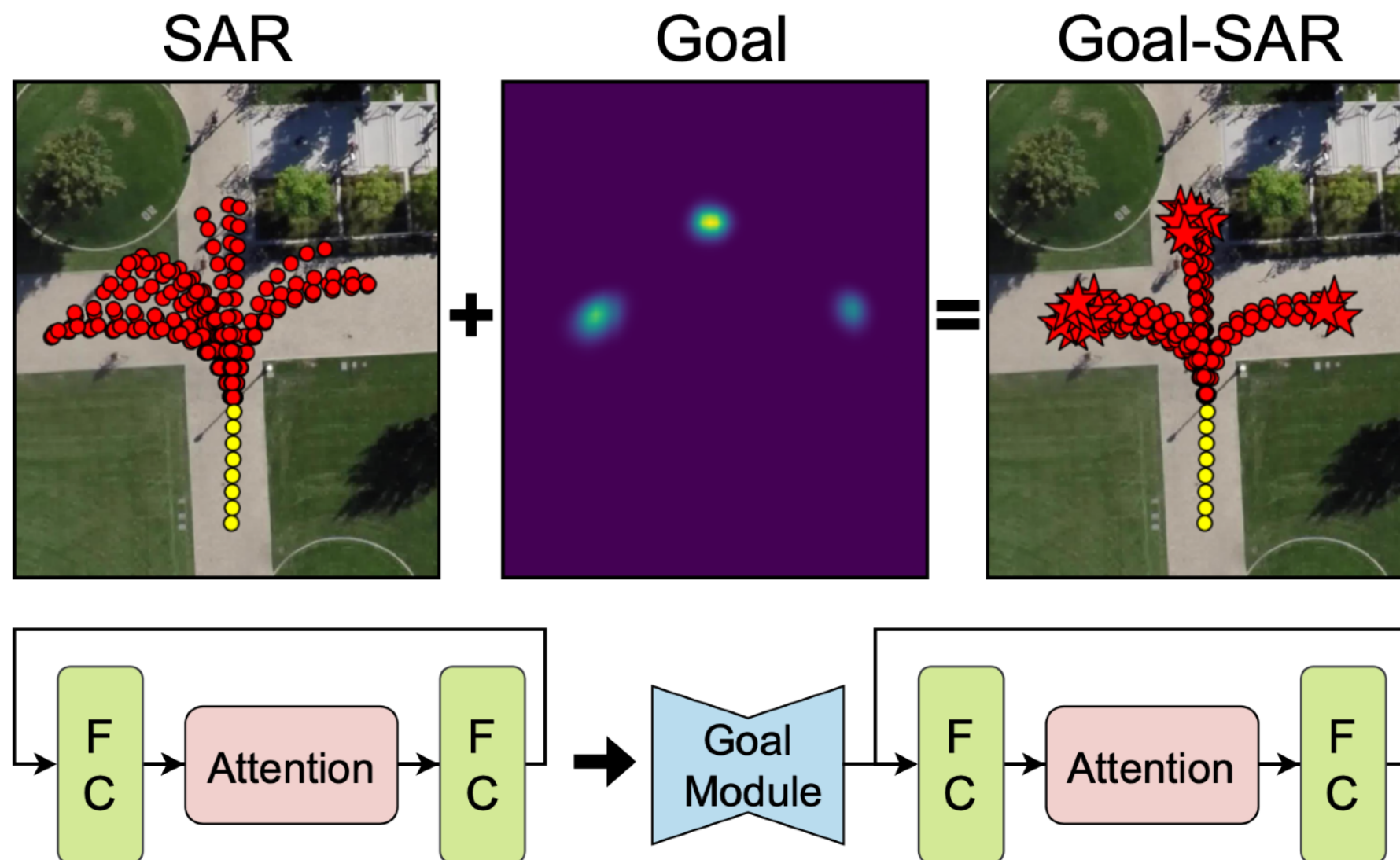
Experiments

- Quantitative results on ETH/UCY, SDD and IND:

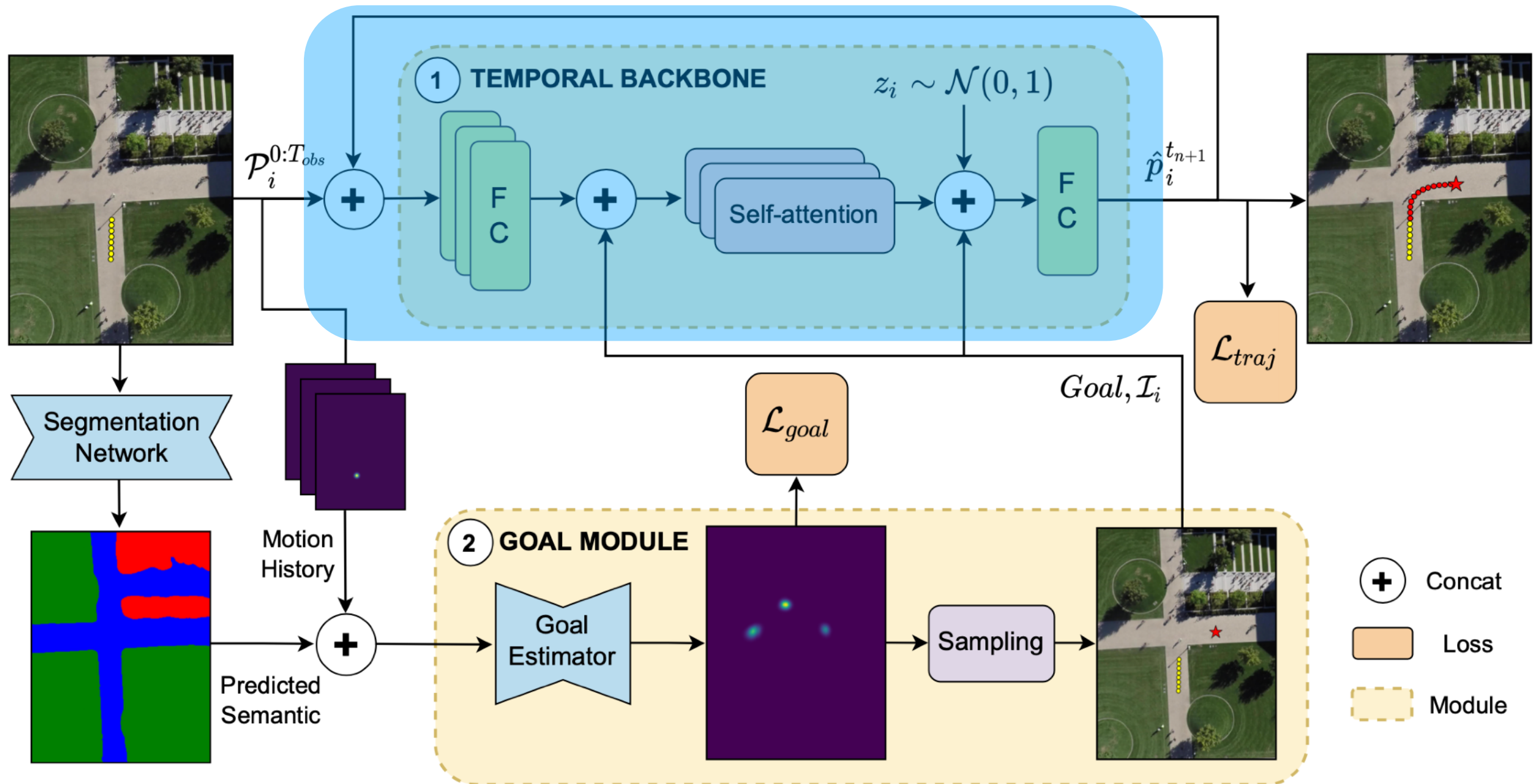
(ADE/FDE)	ETH	HOTEL	UNIV	ZARA1	ZARA2	AVG	SDD	IND
S-LSTM CVPR'16	1.09/2.35	0.79/1.76	0.67/1.40	0.47/1.00	0.56/1.17	0.72/1.54	-	-
S-GAN CVPR'18	0.81/1.52	0.72/1.61	0.60/1.26	0.34/0.69	0.42/0.84	0.58/1.18	0.65/1.26	0.48/0.99
Trajectron ICCV'19	0.59/1.14	0.35/0.66	0.54/1.13	0.43/0.83	0.43/0.85	0.56/1.14	-	-
SoPhie CVPR'19	0.70/1.43	0.76/1.67	0.30/0.63	0.38/0.78	0.54/1.24	0.54/1.15	-	-
STGAT ICCV'19	0.78/1.60	0.30/0.54	0.51/1.08	0.33/0.72	0.29/0.63	0.44/0.91	0.57/1.09	0.48/1.00
AC-VRNN (ours)	0.61/ 1.09	0.30/0.55	0.58/1.22	0.34/ 0.68	0.28/ 0.59	0.42/0.83	0.51/0.90	0.42/0.80

Goals, waypoints and multi-future

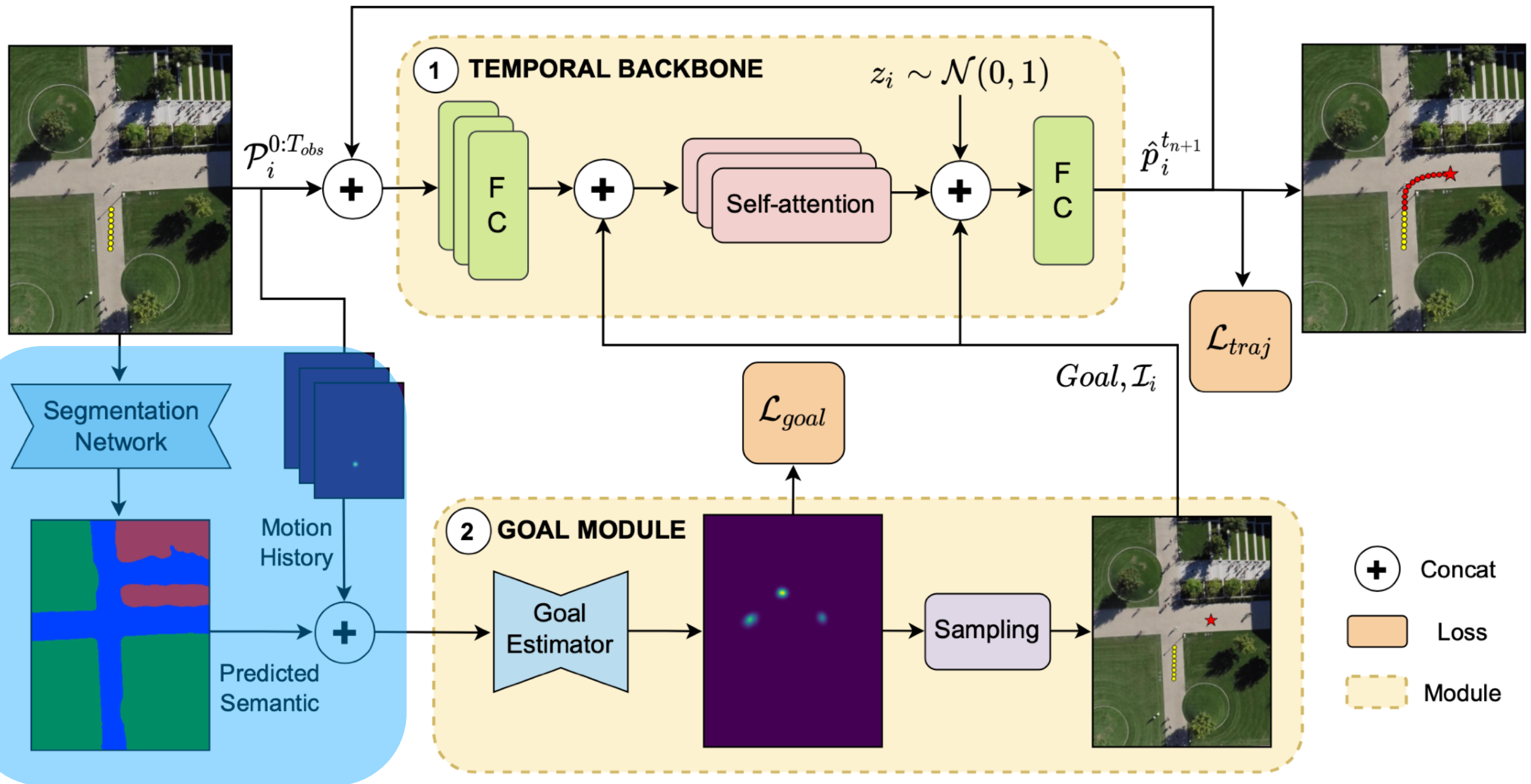
- Goal-driven Self-Attentive Recurrent Networks for Trajectory Prediction



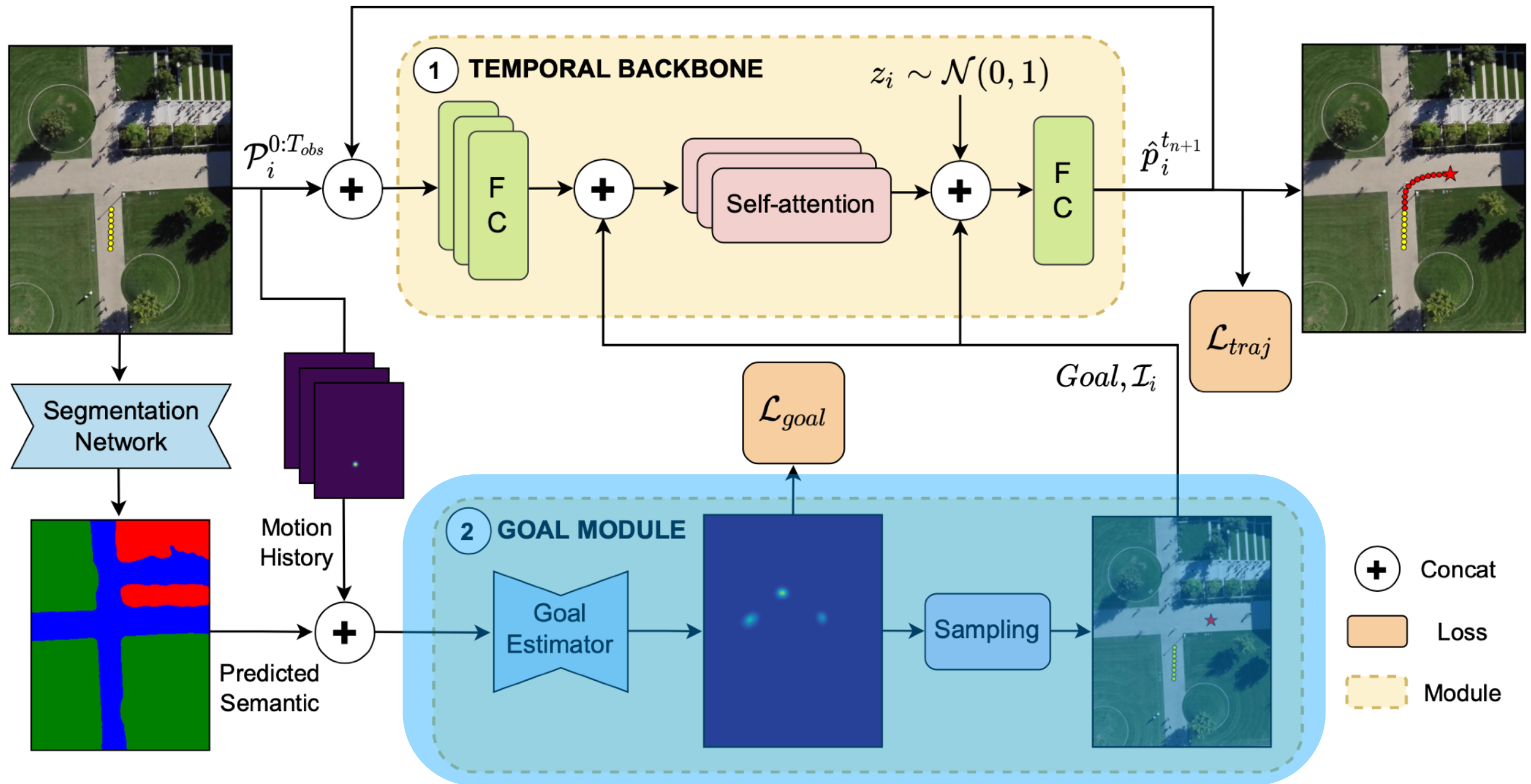
Model architecture



Model architecture



Model architecture



Similarly to: Y-net, ICCV 2021, by Mangalam et al.

Experimental results

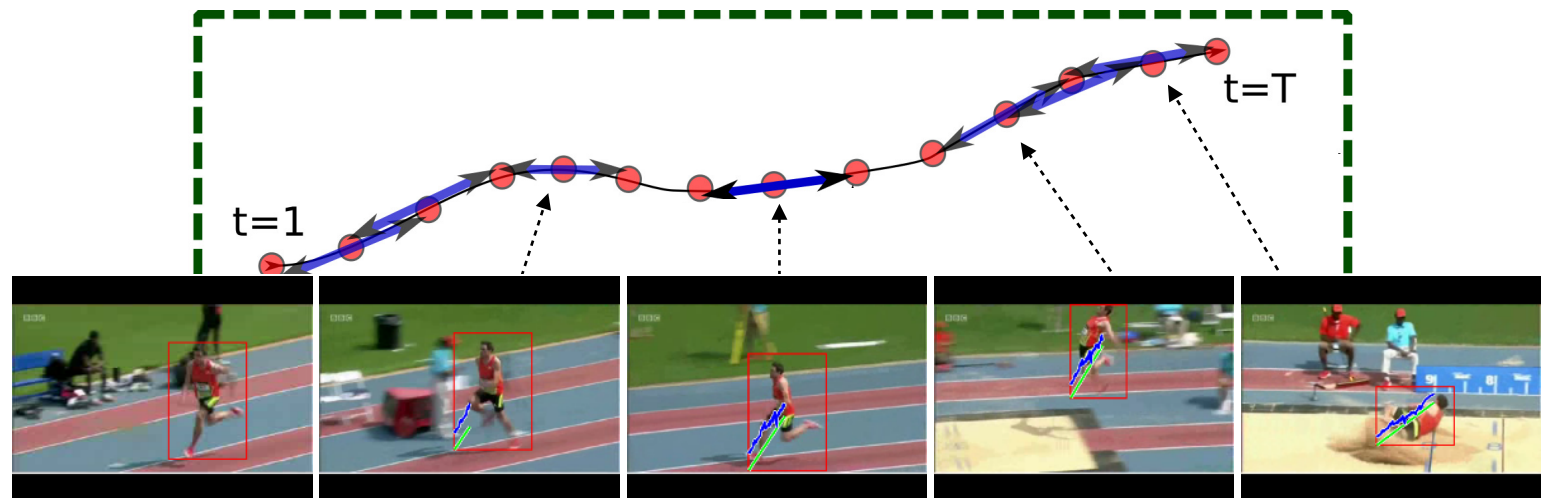
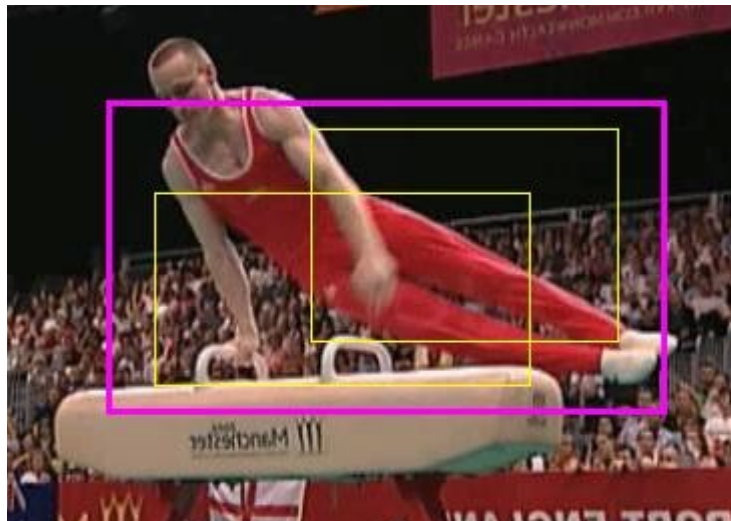
- Standard ETH/UCY benchmark:

Evaluation Metrics (ADE ↓ / FDE ↓) on Min ₂₀									
Method	S	I	G	ETH	HOTEL	UNIV	ZARA1	ZARA2	AVG
Social-LSTM [1]	✓			1.09/2.35	0.79/1.76	0.67/1.40	0.47/1.00	0.56/1.17	0.72/1.54
Social-GAN [14]	✓			0.81/1.52	0.72/1.61	0.60/1.26	0.34/0.69	0.42/0.84	0.58/1.18
ST-GAT [16]	✓			0.65/1.12	0.35/0.66	0.52/1.10	0.34/0.69	0.29/0.60	0.43/0.83
Transformer-TF [13]				0.61/1.12	0.18/0.30	0.35/0.65	0.22/0.38	0.17/0.32	0.31/0.55
STAR [42]	✓			0.36/0.65	0.17/0.36	0.31/0.62	0.26/0.55	0.22/0.46	0.26/0.53
Trajectron++ [35]	✓			0.39/0.83	0.12/0.21	0.20/0.44	0.15/0.33	0.11/0.25	0.19/0.41 ?
AgentFormer [43]	✓			0.26/0.39	0.11/0.14	0.26/0.46	0.15/0.23	0.14/0.24	0.18/0.29
Goal-GAN [10]		✓	✓	0.59/1.18	0.19/0.35	0.60/1.19	0.43/0.87	0.32/0.65	0.43/0.85
PECNet [26]	✓		✓	0.54/0.87	0.18/0.24	0.35/0.60	0.22/0.39	0.17/0.30	0.29/0.48
MG-GAN [9]	✓		✓	0.47/0.91	0.14/0.24	0.54/1.07	0.36/0.73	0.29/0.60	0.36/0.71
Y-net [25]		✓	✓	<u>0.28/0.33</u>	0.10/0.14	<u>0.24/0.41</u>	<u>0.17/0.27</u>	<u>0.13/0.22</u>	0.18/0.27
SAR (Ours)				0.34/0.64	0.14/0.29	0.33/0.66	0.25/0.51	0.21/0.43	0.25/0.51
Goal-SAR (Ours)			✓	<u>0.28/0.39</u>	<u>0.12/0.17</u>	<u>0.25/0.43</u>	<u>0.17/0.26</u>	0.15/0.22	<u>0.19/0.29</u>

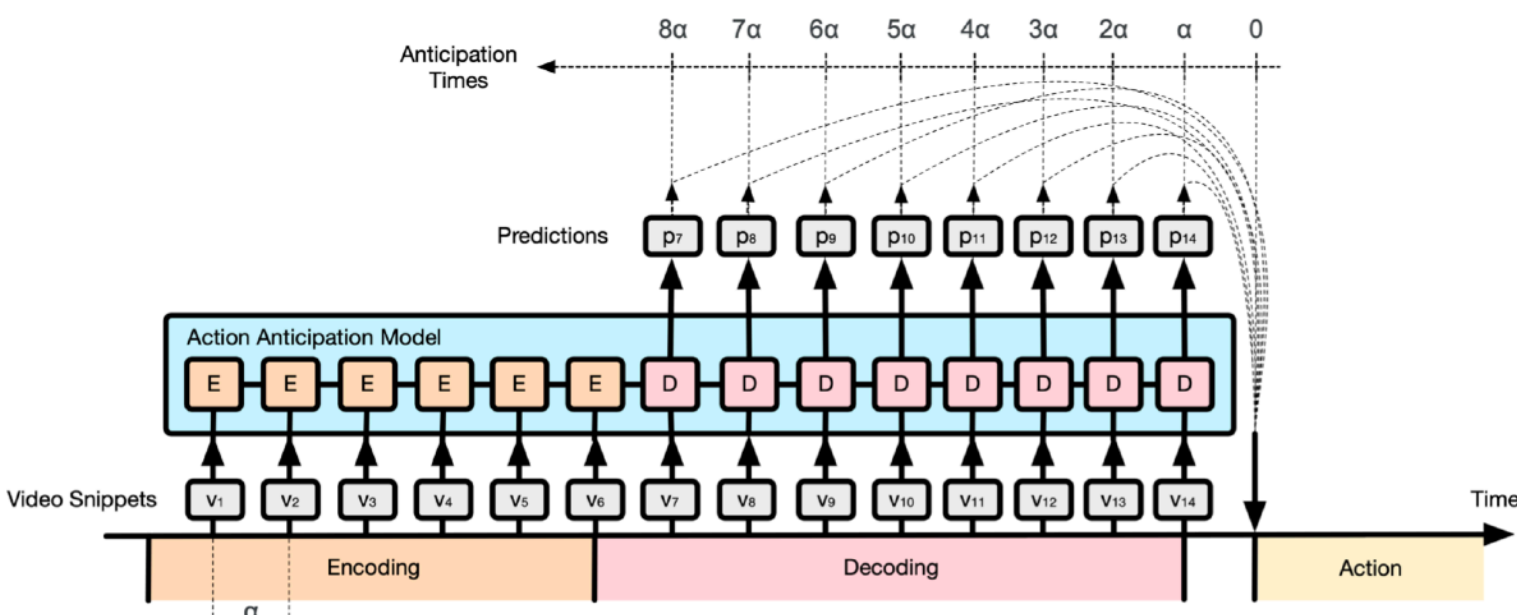
S = social **I** = image/scene data (pixels or semantic) **G** = goal estimate

Predictive vision: other tasks

- Predicting action progress/completion in videos



- Action anticipation and early action detection



EPIC KITCHENS

55 hours

2513 actions

(125 verbs, 352 nouns)

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